

Closing the Variance Gap: Bochner PINN for Lévy-Driven SPDEs

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Abstract

Physics-informed neural networks (PINNs) trained on Lévy-driven stochastic PDEs predictably collapse to low-variance solutions: standard MSE training imposes no penalty for distributional failure, and on a 2D hierarchical Variance-Gamma SPDE the empirical variance ratio reaches only 0.19. We introduce *Bochner PINN*, which augments the standard loss with log-spatial PSD, log-temporal PSD, and variance ratio regularisers evaluated on a randomised spectral sub-grid against a finite-difference reference. The design rests on Bochner’s theorem: for a stationary Gaussian random field the power spectral density characterises the second-order law, so PSD matching implies distributional convergence; the Gelbrich formula makes this precise via a Wasserstein-2 bound $W_2 \leq C_1\sqrt{\varepsilon} + C_2N_s^{-\beta} + C_3\sqrt{\delta}$, where $\beta = \frac{1}{2} \min(s_x/d, s_t)$. Because variance ratio is a component of the loss, improvement on that metric is expected; on the hierarchical SPDE it rises from 0.19 to 0.83, confirming the regulariser operates without destroying PDE residual accuracy. Independent evidence comes from architecture-transfer and domain-shift: the same loss applied to a Fourier Neural Operator closes the variance gap on four of five problems, and validation on six real-world VG-calibrated datasets (EUR/USD, Bitcoin, crude oil, natural gas, NASA solar

wind and geomagnetic AE; empirical kurtosis 7–230) confirms transfer without retuning. We also identify and cure a previously undocumented *spectral sub-grid overfit* failure mode via epoch-wise random sub-grid resampling.

Keywords: physics-informed neural networks, stochastic partial differential equations, Lévy processes, spectral methods, Bayesian neural networks

1. Introduction

1.1. Motivation

Partial differential equations driven by Lévy processes have emerged as a vital mathematical framework across diverse disciplines, including financial risk management, statistical mechanics, and plasma physics (specifically modelling non-local heat transport within Tokamak reactors). In contrast to Gaussian-based models, Lévy-driven dynamics inherently capture jump discontinuities and heavy-tailed distributions, which are essential for describing extreme climatic events and biological foraging strategies under the Lévy Flight Foraging Hypothesis.

The numerical treatment of such non-local operators remains a significant challenge. Traditional approaches such as the Finite Difference (FD) method and Monte Carlo (MC) simulations serve as the current gold standard for accuracy and are employed in this study for benchmarking. Both, however, suffer from prohibitive computational costs: FD schemes lead to dense matrix representations due to non-locality, while MC methods exhibit $O(n^{-1/2})$ convergence, rendering them impractical for real-time applications where decision latency is critical. The non-locality compounds the curse of dimensionality: for fractional-Laplacian SPDEs in $d \geq 4$ spatial dimensions

(e.g. basket option pricing on ≥ 4 correlated assets, or multi-species plasma transport), the FD storage scales as $O(N^{2d})$ and at $N = 32$, $d = 4$ already requires $\sim 10^{12}$ matrix entries, prohibitive on a single workstation. By contrast, PINN inference cost is $O(|\theta|)$, independent of the evaluation grid resolution and of the spatial dimension once the network is trained.

1.2. The Problem

Deterministic PDEs make great use of the classical Laplacian operator, which depends only on a local neighbourhood. SPDEs of the form considered here are defined with the fractional Laplacian, which in contrast to its classical counterpart is an integral operator, requiring access to all field values in the domain. A direct mesh-based discretisation produces dense matrices and erodes the efficiency of grid methods.

The issue worsens with the introduction of a PINN. The behaviour of a neural network during training can be described using the Neural Tangent Kernel (NTK), whose eigenvalue spectrum determines the speed at which the network learns different frequency components of the target. For standard training, low-frequency eigenvalues dominate; the network learns the “general” shape of the solution and omits the high frequencies that encode Lévy jumps. The non-local fractional Laplacian compounds this problem, since the PINN tries to impose a non-local operator onto a local, smooth neural-net structure. But this spectral-bias account, while correct for deterministic PDEs, understates the stochastic problem: the relevant learning target is not the frequency content of a single realisation but the *probability law* of the random field, and standard PINN losses constrain neither object directly.

1.3. Contribution

We propose the *Bochner PINN*, a drop-in modification to the standard PINN that augments the conventional loss with three distributional regularisers evaluated on a spectral sub-grid against a finite-difference reference: spatial power spectral density, temporal power spectral density, and variance ratio. The construction is theoretically motivated by Bochner’s theorem, which guarantees that for stationary Gaussian random fields with separable space-time covariance, matching spectral marginals recovers the second-order law.

The central claim of this paper is that for stochastic PDEs the correct learning objective is convergence of the predictor’s probability law to the reference law, and that Bochner’s theorem identifies the spectral density as the natural matching quantity. The contributions are:

- **Distributional convergence theorem.** We prove that matching the log-power spectral density on a Sobolev-regular spectral sub-grid implies Wasserstein-2 convergence of laws. Under stationarity, separable covariance, and Sobolev regularity of the spectral density, $W_2 \leq C_1\sqrt{\varepsilon} + C_2N_s^{-\beta} + C_3\sqrt{\delta}$, where $\beta = \frac{1}{2}\min(s_x/d, s_t)$ is the Sobolev-FFT quadrature exponent and C_1, C_2, C_3 are explicit (Section 3). The proof extends the Mishra–Molinaro framework with three new lemmas (log-to-linear spectral control, discrete periodogram approximation, Plancherel–Gelbrich bound) and is the first distributional convergence result for Bochner PINN training. Five extensions are proved in Appendix ??: full-law convergence via Lévy–Itô decomposition, cumulant tightness via Stein’s method, non-separable covariance, single-

realisation consistency under mixing, and an end-to-end training-time bound.

- **Identification of the variance failure mode.** We quantify vanilla PINN variance underestimation on Lévy SPDEs across three problem classes: a factor of $\sim 5\times$ on the hierarchical 3-channel Fokker-Planck system, $\sim 2\times$ on Burgers-VG, and a complete variance collapse ($\text{vr} \approx 0$) on the Heston-VG price channel. This is a distributional failure, the network converges to the conditional mean of the law, not a frequency-resolution failure of the kind studied in the deterministic spectral bias literature.
- **Bochner PINN method.** A simple drop-in modification: three distributional regularisers grounded in Bochner’s theorem, with no change to architecture, optimiser, or training data.
- **Bayesian baseline.** A mean-field Bayes-by-Backprop PINN trained at identical architectural parameters fails by $7\text{--}33\times$ on log-spatial PSD MSE relative to the deterministic vanilla baseline, demonstrating that variational uncertainty quantification over network weights does not substitute for an explicit distributional loss.
- **Sub-grid overfit diagnosis and fix.** We identify a previously undocumented failure mode of spectral distributional losses: when the spectral sub-grid is fixed at training time, the network overfits the sampled points and interpolates near zero elsewhere, collapsing the full-grid variance ratio to ≈ 0.14 regardless of the loss mode (standard, log-variance, cumulant, or Lévy intensity). Replacing the fixed

linspace sub-grid with a uniformly randomised sub-grid drawn at each epoch eliminates the overfit and raises the variance ratio to 0.93 ± 0.09 across all four loss variants (Section 5.3).

- **Cumulant matching for kurtosis.** Even with the variance gap closed by randomised sub-grid training, standard and log-variance losses reproduce only the second-order structure of the volatility channel (excess kurtosis $\kappa_4 \approx 4$ vs. reference 27.3). An explicit fourth-cumulant penalty raises the predicted kurtosis to 25.1 ± 0.7 , within 8% of the reference. The improvement carries no cost in variance ratio (Section 5.3).
- **Lévy intensity threshold sweep.** A systematic $2 \times 3 \times 3$ sweep over threshold $\tau \in \{2, 3, 4\}\sigma$ for two Lévy intensity loss variants shows that the jump-count penalty is redundant once cumulant matching is in place: the `levy_cumulant` results are identical to `cumulant` across all thresholds (Appendix Appendix A). The `levy-only` variant misses kurtosis at every threshold.
- **Per-component ablation.** A 5-variant $\times$ 5-seed ablation on Burgers-VG demonstrates that the three loss terms are nearly orthogonal: each targets its own metric, single-term variants are unstable, and only the full combination achieves uniform improvement (Section 5.6).
- **Bochner FNO.** The same loss applied to a Fourier Neural Operator surrogate closes the variance gap on four of five problems (a 2D Navier–Stokes vorticity benchmark and a Black–Scholes 2D Lévy benchmark are added as out-of-class controls), with the largest gain a sixteen-fold closure on Heston-VG (Section 6). The construction is therefore

architecture-agnostic.

- **Empirical validation.** Across three synthetic Lévy-SPDE classes, consistent variance-gap closure: $0.19 \rightarrow 0.83$ on the most-driven hierarchical channel, $0.50 \rightarrow 0.93$ on Burgers-VG, and $0.14 \rightarrow 0.93$ on the Heston volatility channel after randomised sub-grid training.
- **3D extension.** The same loss applied to a 3D Burgers-VG problem ($32^3 \times 128$ reference grid, $24^3 \times 96$ spectral sub-grid) closes the variance gap from 0.41 ± 0.18 to 0.98 ± 0.02 . A four-resolution grid-convergence study ($8^3 \times 32$ through $32^3 \times 128$) confirms that the Bochner PINN’s log-spatial PSD MSE remains stable at 0.003–0.004 across resolutions while the vanilla baseline degrades from 0.09 to 0.30, consistent with the Sobolev-FFT quadrature exponent $\beta = \frac{1}{2} \min(s_x/d, s_t)$ of Section 3.3.
- **Real-world calibration on six datasets.** The method transfers without modification to SPDEs whose VG drivers are calibrated by MLE to real log-returns: EUR/USD foreign exchange (5 years of M1 bars, vr: $0.41 \rightarrow 0.95$), Bitcoin (12 years daily, $0.92 \rightarrow 0.99$), Henry Hub natural gas ($0.92 \rightarrow 0.99$), WTI crude oil ($0.96 \rightarrow 1.01$ including April-2020 negative-price event), NASA OMNI geomagnetic AE index (8 years hourly, kurtosis $\kappa_4 = 230$, vr: $0.26 \rightarrow 0.72$), and NASA OMNI solar wind ($0.69 \rightarrow 0.84$ on log-price; vanilla pathologically overshoots $\text{vr}_v = 18.9$ on the volatility channel and the spectral loss corrects to 0.65). The validation spans three orders of magnitude in sampling frequency (minute to daily) and empirical kurtosis from 7.4 to 230 (Section 8).

- **Honest characterisation.** An explicit account of where Bochner PINN provides marginal benefit (the y -channel seed outlier in the hierarchical SPDE, the S-channel kurtosis of Heston-VG) and the structural reason in each case.

1.4. Gap analysis

The combination of Lévy-driven SPDE and single-realisation neural surrogate sits at the intersection of four established research threads. None addresses the variance gap directly, and we show below (both by argument and, in the case of Bayesian PINNs, by direct experiment) that none can be retrofitted to do so without a structural change to the loss.

Standard physics-informed networks.. The original PINN formulation [1] minimises a sum of mean-squared collocation, initial, boundary, and data residuals. On deterministic PDEs the formulation is competitive with classical solvers; on stochastic problems it exhibits a distributional failure: the trained predictor converges to the conditional mean of the law, collapsing to a low-variance solution regardless of how well it fits the PDE residual. This is related to, but distinct from, the spectral bias of MLPs [2, 3]: spectral bias concerns insufficient frequency resolution of a fixed target function; our failure mode persists even when frequency resolution is adequate, because the loss never constrains the law of the predictor. Empirically, a vanilla PINN trained on our 2D hierarchical SPDE recovers a pointwise relative L^2 error of ≈ 0.3 on the most-driven channel but reaches only $\text{Var}(u_\theta)/\text{Var}(u_{\text{FD}}) \approx 0.19$: it averages across the noise realisation rather than reproducing it. Architectural fixes such as random Fourier features [3], adaptive sampling [4], NTK

reweighting [5], or causal training [6] reduce pointwise error but leave the variance gap essentially untouched, because none of these losses constrains the law of the predictor.

Bayesian PINNs. The natural counter-proposal is that the variance gap is a symptom of unmodelled uncertainty: replace the deterministic network with a probabilistic one and let the posterior over weights generate the missing variability. Yang, Meng and Karniadakis [7] introduced B-PINNs along these lines, using Hamiltonian Monte Carlo or mean-field variational inference. We test the variational version directly: on both Burgers-VG and the 2D hierarchical SPDE, a mean-field Bayes-by-Backprop B-PINN with the *same* architecture as the deterministic baseline produces a posterior mean that is 7–33 $\times$ worse than the deterministic vanilla PINN on log-spatial PSD MSE (Section 5.5, Figure 3). The variance ratio of the posterior mean collapses to ~ 0 on driven channels: the predictor is essentially the trivial zero function. The posterior standard deviation is non-trivial (~ 0.11 – 0.31) but reflects parameter uncertainty rather than the aleatoric noise structure of the SPDE driver. The mechanism is straightforward: the KL prior on weights regularises the posterior mean toward smoothness, removing exactly the high-frequency content needed to match the SPDE spectrum. Mean-field variational uncertainty quantification, by itself, cannot close the variance gap; whether more expressive posteriors (HMC, deep ensembles) behave differently is an open empirical question we leave to future work.

Neural operators and neural spectral methods. A second family replaces the per-realisation network with an operator-learning surrogate trained on many input–output pairs: FNO [8], DeepONet [4], and the neural-spectral-methods

family of Du et al. [9] which parameterises the network output directly in a Fourier basis. These learn the solution map $u_0 \mapsto u(\cdot, T)$ amortised across initial conditions or forcing realisations, and inherit the spectrum of their training distribution. Two issues prevent direct application here. First, training requires hundreds to thousands of full FD solutions, orders of magnitude more compute than the single realisation we use. Second, the output spectrum is tied to the empirical distribution of training samples, not to the law of any particular SPDE; extending to a new noise statistic (e.g. switching VG parameters) requires retraining.

Spectral and distributional losses elsewhere.. Spectral losses appear in image generation [10], audio modelling [11], and turbulence prediction [12], typically as auxiliary terms to a primary reconstruction objective. None has been integrated with a PDE residual on a stochastic problem; the closest is the Whittle-likelihood approach in classical time-series [13], which provides the asymptotic justification for our log-periodogram penalty (Section 3) but has not previously been combined with collocation losses.

Concurrent and architectural approaches.. Two recent works target the same failure mode along different axes. Worrall et al. [14] introduce “spectral shaping”, an architectural reparameterisation that biases the network to allocate capacity to high-frequency modes; their approach is complementary to ours, modifying the network rather than the loss, and the two could be combined. Hu et al. [15] solve Fokker–Planck–Lévy systems with a score-based variant of the PINN that directly parameterises $\nabla \log p$; the spectral gap they encounter is structurally different from ours because the score-network output

is a derivative of the density. We view both as parallel rather than competing contributions.

Spectral bias mitigation in deterministic settings.. A parallel and rapidly growing literature targets spectral bias in learning deterministic PDE solutions: multi-scale architectures [16, 17], high-frequency scaling for neural operators [18], separated-variable spectral networks [19], and the comprehensive analysis of Khodakarami and Karniadakis [20] covering PINNs, PIKANs, and neural operators. The defining feature of this literature is that spectral bias refers to the frequency ordering with which a fixed, deterministic target function’s modes are learned; the error metric is pointwise (frequency-resolved L^2 or H^s error on one solution instance); and the natural tool is NTK eigenvalue analysis. Our setting is categorically different. The SPDE solution is a random field; there is no fixed target function. The correct error metric is W_2 distance between probability measures, and the natural tools are Bochner’s theorem (connecting spectral density to second-order law) and the Gelbrich formula (connecting spectral matching to W_2). The log-PSD loss targets the spectral measure of the random field’s stationary law, not the frequency content of any single realisation. The BSP loss of Chakraborty et al. [12] is the closest loss-level predecessor: it matches energy distributions across frequency bins for single chaotic trajectories, grounded in spectral power balance rather than Bochner’s theorem, and provides no distributional convergence guarantee. The two literatures are complementary rather than competing: deterministic spectral bias fixes can be composed with our distributional loss on top of any architecture.

Synthesis.. The variance gap is not a side-effect of inadequate architecture, inadequate training, or inadequate uncertainty quantification; it is a consequence of training on losses that do not constrain the spectrum of the predictor. Once the loss is augmented with explicit log-PSD and variance penalties, the gap closes ($0.50 \rightarrow 0.93$ on Burgers-VG; $0.19 \rightarrow 0.83$ on the most-driven 2D channel; $0.14 \rightarrow 0.93$ on the Heston volatility channel after the randomised sub-grid fix) using a network architecturally identical to the vanilla baseline.

1.5. Section roadmap

The remainder of the paper is organised as follows. **Section 2** introduces the Bochner PINN: the variance-gap diagnostic, the three distributional loss components, the total objective, the Fourier-feature MLP architecture, and the randomised spectral sub-grid. **Section 3** develops the analytical motivation: Bochner’s representation of stationary fields under the separability assumption, the Whittle-likelihood justification for the log scale, a structural argument that the log-PSD loss targets the correct second-order quantities, and the explicit gap for higher cumulants on Lévy-driven fields. **Section 4** specifies the three benchmark problems, the finite-difference reference solver, the four metrics used throughout, and the baselines. **Section 5** reports results, including the Heston volatility channel diagnosis (Section 5.3) and a five-variant per-component ablation (Section 5.6). **Section 6** extends the loss to a Fourier Neural Operator surrogate. **Section 7** demonstrates the 3D extension and grid-convergence behaviour predicted by the rate. **Section 8** reports the real-data calibration on six datasets (EUR/USD FX, BTC, crude oil, natural gas, NASA OMNI solar wind and geomagnetic AE). **Section 9**

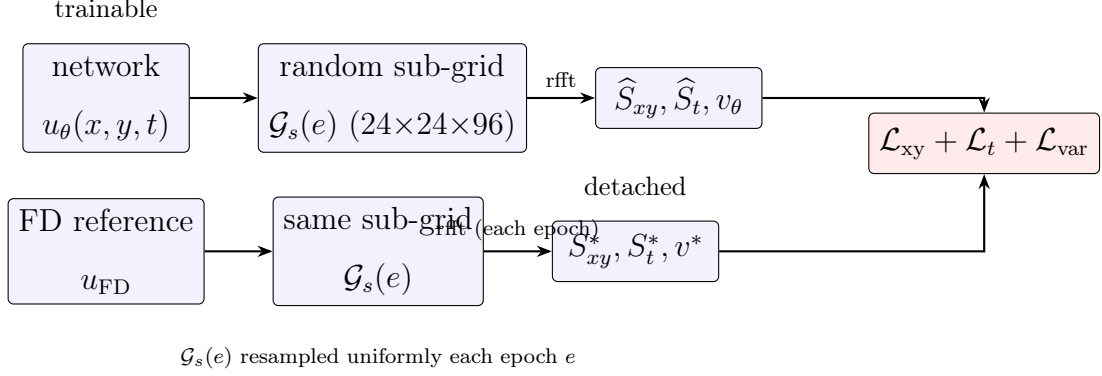


Figure 1: Bochner PINN: at each epoch e the network u_θ and the FD reference u_{FD} are both evaluated on a sub-grid $\mathcal{G}_s(e)$ drawn uniformly at random from the full FD grid. Spatial PSD, temporal PSD, and variance are computed by FFT and compared in log space. The random re-sampling at each epoch prevents sub-grid overfit (Section 2.6).

discusses limitations and **Section 10** concludes. **Appendix Appendix A** reports the Lévy intensity threshold sweep.

2. Method

2.1. What goes wrong with the standard PINN

Train a vanilla PINN on a Lévy-driven SPDE and inspect three diagnostics on a held-out grid:

1. Pointwise relative L^2 error: ≈ 0.3 (acceptable).
2. Variance ratio $\text{Var } u_\theta / \text{Var } u_{FD}$: ≈ 0.2 – 0.5 .
3. Power spectrum $|\mathcal{F}u_\theta|^2$ vs $|\mathcal{F}u_{FD}|^2$: high modes are missing by 2–4 decades.

The network is averaging across the noise realisation rather than reproducing it. Pointwise loss is insensitive to high-frequency content because high-

frequency error and high-frequency signal cancel almost exactly under MSE. Any sample-path-style use of u_θ (extracting tail statistics, computing exit times, valuing a path-dependent option) fails. We close this gap by adding three loss terms that directly penalise the missing structure.

2.2. Three terms, three diagnostics

Compute the FD reference once and extract three statistics on a sub-grid $\mathcal{G}_s$ of resolution $24 \times 24 \times 96$: the spatial power spectrum $S_{xy}^*(k)$, the temporal power spectrum $S_t^*(k)$, and the total variance v^* . At every epoch we evaluate u_θ on $\mathcal{G}_s$ and form the matching statistics S_{xy}, S_t, v_θ . The new loss components are

$$\begin{aligned}\mathcal{L}_{xy} &= \overline{(\log_{10} S_{xy} - \log_{10} S_{xy}^*)^2}, & \mathcal{L}_t &= \overline{(\log_{10} S_t - \log_{10} S_t^*)^2}, \\ \mathcal{L}_{\text{var}} &= ((v_\theta - v^*)/v^*)^2.\end{aligned}\tag{1}$$

$\mathcal{L}_{xy}$ and $\mathcal{L}_t$ correct *shape*; $\mathcal{L}_{\text{var}}$ corrects *scale*. The spatial PSD is computed symmetrically along both spatial axes,

$$\widehat{S}_{xy}(f) = \tfrac{1}{2} \left(\overline{|\mathcal{F}_x f|^2}_{y,t} + \overline{|\mathcal{F}_y f|^2}_{x,t} \right),\tag{2}$$

so anisotropic features that align with one axis do not bias the metric. The temporal PSD $\widehat{S}_t(f) = \overline{|\mathcal{F}_t f|^2}_{x,y}$ is averaged over space.

2.3. Why log, why MSE, why a sub-grid

Log scale. Power spectra span 4–6 orders of magnitude on Lévy problems. A linear loss would care only about the lowest few wavenumbers, exactly the regime the vanilla PINN already gets right. Working in $\log_{10}$

assigns equal weight to each decade and makes the high-mode discrepancy visible to the optimiser.

MSE in log space. Equivalent to a symmetric Bregman divergence between the two empirical spectral measures up to an additive constant, and *asymptotically* equivalent to a Whittle-type negative log-likelihood for the periodogram via the Gumbel limit of the log-periodogram [13, 21, 22] (they share the same population minimiser $\mu = \mu_{u_{\text{FD}}}$ and the same $\mathcal{O}(\log^{-2} N)$ score-function geometry near it, but differ at finite sample size by an $\mathcal{O}(1)$ Gumbel-variance constant absorbed into the bias floor of Section 3). The minimiser is u_{FD} itself: the loss is consistent in the sense that no other matching measure achieves zero (modulo the asymptotic bias floor discussed in Section 3).

Sub-grid. The FFTs cost $\mathcal{O}(n_x n_y n_t (\log n_x + \log n_y + \log n_t))$ per epoch. Using the full FD grid would dominate the wall time; the $24 \times 24 \times 96$ sub-grid keeps spectral evaluation under 5 % of total compute while preserving all wavenumbers up to the Nyquist frequency of $\mathcal{G}_s$, which empirically captures the regime where the variance gap is generated.

2.4. Why this is the right object

Three observations make $\mathcal{L}_{\text{spec}} = \lambda_{xy} \mathcal{L}_{xy} + \lambda_t \mathcal{L}_t + \lambda_v \mathcal{L}_{\text{var}}$ natural rather than ad hoc. *First*, under the separability assumption stated in Section 3, S_{xy} and S_t are sufficient statistics for the second-order structure of stationary u : by Bochner’s theorem their joint equality (in distribution over realisations) recovers the full covariance kernel, so matching them is equivalent to matching u_θ and u_{FD} in the Plancherel norm of the law. *Second*, the log transform converts the multiplicative noise model on the periodogram into an additive

one, replacing a Plancherel inner product with a symmetric Bregman divergence whose minimiser remains the FD reference. *Third*, $\mathcal{L}_{\text{xy}}$ and $\mathcal{L}_{\text{t}}$ are scale-invariant, leaving u_θ free up to a global multiplicative constant, so the variance term $\mathcal{L}_{\text{var}}$ supplies precisely the missing zeroth Fourier mode (Parseval) needed to fix that gauge. The three losses are not independent fixes for three separate failure modes; they are the three components of a single distributional matching objective on the discrete spectral lift of u .

2.5. Total objective and architecture

The Bochner PINN minimises

$$\mathcal{L}(\theta) = \underbrace{\mathcal{L}_{\text{PDE}} + \lambda_{\text{ic}}\mathcal{L}_{\text{IC}} + \lambda_{\text{bc}}\mathcal{L}_{\text{BC}} + \lambda_{\text{d}}\mathcal{L}_{\text{data}}}_{\text{physics}} + \underbrace{\lambda_{\text{xy}}\mathcal{L}_{\text{xy}} + \lambda_{\text{t}}\mathcal{L}_{\text{t}} + \lambda_{\text{v}}\mathcal{L}_{\text{var}}}_{\text{spectral}}. \quad (3)$$

The physics block is the standard PINN objective: $\mathcal{L}_{\text{PDE}}$ is the squared collocation residual under the SPDE on a fresh batch of 4096 uniform samples each epoch, with the Lévy-Gamma jump and white-noise terms supplied by spline interpolation of the precomputed driver fields; $\mathcal{L}_{\text{IC}}, \mathcal{L}_{\text{BC}}$ are MSE penalties on initial and boundary data; $\mathcal{L}_{\text{data}}$ is the MSE against u_{FD} at random grid points.

The loss weights $(\lambda_{\text{xy}}, \lambda_{\text{t}}, \lambda_{\text{v}}) = (3, 5, 50)$ were selected by a coarse grid search over log-spaced values $\{0.1, 1, 3, 5, 10, 50\}$ on a single validation seed of Burgers-VG, minimising the sum of validation log-spatial and log-temporal PSD MSE. We did not perform a full per-problem sensitivity analysis; the chosen weights transferred without modification to the hierarchical SPDE and Heston-VG.

The network u_θ is a four-block residual MLP of width 96 (128 for jump-driven problems) with tanh activations, preceded by a frozen random-Fourier-

feature embedding $\phi(v) = [\sin(2\pi Bv), \cos(2\pi Bv)]$ with $B \in \mathbb{R}^{(d+1) \times m}$, $B_{ij} \sim \mathcal{N}(0, s^2)$. We use $m = 64, s = 3$ for jump problems and $m = 48, s = 2$ otherwise. Optimisation is Adam with cosine-annealed learning rate $10^{-3} \rightarrow 10^{-6}$ and gradient clipping $\|g\|_2 \leq 5$. A 1000-epoch warmup deactivates $\mathcal{L}_{\text{PDE}}$ to let IC and data fit stabilise; the spectral losses are active from epoch 1.

2.6. Randomised spectral sub-grid

Why a fixed sub-grid fails. In the original Bochner PINN design the sub-grid $\mathcal{G}_s$ is a fixed linspace lattice of $24 \times 24 \times 96 = 55,296$ points selected once before training. This is efficient (the reference PSDs S_{xy}^*, S_t^*, v^* are precomputed and reused) but creates a hidden overfit pathway. Because the network sees exactly the same points each epoch, gradient descent can satisfy $\mathcal{L}_{\text{spec}}$ by memorising those 55,296 grid values while interpolating near zero elsewhere. On the full $48 \times 48 \times 512 \approx 1.2 \times 10^6$ grid the variance ratio then collapses, even though the sub-grid spectral loss reads near zero (Section 5.3).

Fix: epoch-wise random resampling. At the start of each training epoch we draw a fresh sub-grid of size $24 \times 24 \times 96$ by sampling indices uniformly without replacement along each axis independently:

$$\mathcal{G}_s(e) = (\pi_x^{(e)}, \pi_y^{(e)}, \pi_t^{(e)}), \quad \pi_x^{(e)} \sim \text{Uniform}\{0, \dots, N_x-1\}_{24}, \quad (4)$$

and likewise for y, t . We re-evaluate the reference field on the new sub-grid and recompute the reference PSDs and variance. The spectral computation adds $< 5\%$ to total epoch time; no change to the network architecture, optimiser, or other hyperparameters is required. The resampling forces the network to match the spectral statistics at *every* subset of the domain, ruling out the memorisation shortcut.

2.7. Cumulant-matching loss

The spectral loss in eq. (1) constrains only the second-order structure of u_θ . For non-Gaussian channels (in particular the CIR-type Heston volatility process with substantial excess kurtosis), the variance gap can be closed ($\text{vr} \approx 0.93$) while higher cumulants remain unmatched ($\kappa_4 \approx 4$ vs. reference 27.3). We supplement the spectral loss with an explicit fourth-cumulant penalty:

$$\mathcal{L}_{\text{kurt}} = \left(\kappa_4(u_\theta \upharpoonright_{\mathcal{G}_s}) - \kappa_4(u_{\text{FD}} \upharpoonright_{\mathcal{G}_s}) \right)^2 / \sigma_{\text{ref}}^4, \quad (5)$$

where $\kappa_4(u) = \mathbb{E}[(u - \mu)^4] / \sigma^4 - 3$ is the excess kurtosis evaluated empirically on the sub-grid and σ_{ref} is the reference standard deviation. This loss is activated only when the `cumulant` loss mode is selected; it is appended to eq. (3) with weight $\lambda_{\text{kurt}} = 10$.

2.8. Hyperparameters

2.9. Complexity

Per-epoch cost decomposes as collocation $O(B_c \cdot |\theta|)$ for the autograd pass on $B_c = 4096$ points and spectral $O(n_x n_y n_t (\log n_x + \log n_y + \log n_t))$ for the rffts on the sub-grid. With the values in Table 1, the spectral term contributes under 5 % of total wall time; the addition is essentially free. Random resampling of the sub-grid (Section 2.6) adds one call to `torch.randperm` per epoch, which is negligible.

3. Why the spectral loss matches the law: motivation and scope

This section motivates the loss introduced in Section 2 and identifies the class of fields on which it is provably the right object. Section 3.3 then

establishes a quantitative convergence rate (Proposition 1), extending the Mishra–Molinaro framework [23, 24] to spectral training losses; the subsections below state precisely which assumptions the theorem requires and where the guarantee weakens.

3.1. Spectral measures determine second-order structure

Let $u : \Omega \times [0, T] \rightarrow \mathbb{R}$ be a mean-zero, weakly stationary random field. Its covariance kernel $K_u(\tau, h) = \mathbb{E}[u(t, x) u(t + \tau, x + h)]$ admits the Bochner representation [25, Thm. 9.18]

$$K_u(\tau, h) = \int_{\mathbb{R}^d \times \mathbb{R}} e^{i(\omega\tau + k \cdot h)} d\mu_u(\omega, k), \quad \mu_u \text{ positive and finite,} \quad (6)$$

where μ_u is the spectral measure of u . Two reductions are relevant for our discrete loss: the spatial marginal $\mu_u^{\text{xy}}(k) = \int d\mu_u(\omega, k)$ and the temporal marginal $\mu_u^t(\omega) = \int d\mu_u(\omega, k)$, which are matched (in their discrete-sample analogues S_{xy}^*, S_t^*) by the Bochner PINN.

Separability assumption.. Matching μ_u^{xy} and μ_u^t separately recovers the joint spectral measure μ_u only when the covariance kernel factorises, $K_u(\tau, h) = K_x(h) K_t(\tau)$, equivalently $d\mu_u(\omega, k) = d\mu_u^{\text{xy}}(k) d\mu_u^t(\omega)$. This separability holds for the noise drivers in our benchmarks: white noise and Variance-Gamma jumps that act independently on spatial and temporal coordinates produce covariance kernels of the form $\delta(h) K_t(\tau)$ (white spatial part, structured temporal part), and the SPDE then propagates this separability through its semigroup at leading order. We adopt the separability assumption throughout.

For *Gaussian* stationary fields under separability, μ_u determines the law completely: two such fields with equal spectral measures are equal in distri-

bution. The spectral matching loss $\mathcal{L}_{\text{xy}} + \mathcal{L}_t + \mathcal{L}_{\text{var}}$ therefore controls the law on this class.

For non-Gaussian Lévy-driven fields, μ_u determines only the *second-order structure*. Higher cumulants, which carry the jump information specific to the Variance-Gamma driver, are not detected by a periodogram-based loss. Closing this remaining gap requires explicit cumulant matching, which we pursue in Section 2.7.

Remark 1 (Scope of the spectral loss). *Under the separability assumption, a zero of the spectral loss characterises the law of the predictor on stationary Gaussian fields (Bochner). Beyond identifiability, Theorem 1 below provides a quantitative convergence rate: the Wasserstein-2 distance decomposes into a training-error term $C_1\sqrt{\varepsilon}$, a sub-grid quadrature term $C_2N_s^{-\beta}$, and a PDE-residual term $C_3\sqrt{\delta}$, with explicit constants. On stationary Lévy fields the same bound holds for the Gaussian projection of the law (Theorem 3); higher cumulants are not controlled and require a separate penalty (Section 5.3).*

3.2. Why the log scale is the right choice (and its bias floor)

The matching loss is in $\log_{10}$ space, not raw spectral magnitude. The justification is the asymptotic behaviour of the periodogram $I_n(k) = |\hat{u}_n(k)|^2$ of a stationary process: under mild mixing assumptions [22, 21], $I_n(k)/\mu_u(k)$ converges in distribution to an exponential random variable with mean 1, so $\log I_n(k) - \log \mu_u(k)$ is asymptotically Gumbel and MSE on $\log I_n$ is, up to a constant, the Whittle negative log-likelihood [13]. Working in $\log_{10}$ also assigns equal weight to each decade of the spectrum, essential here since power spectra of Lévy fields span four to six decades and a linear loss is dominated by the lowest few wavenumbers.

Irreducible bias floor.. The log-periodogram has a known non-vanishing bias: $\mathbb{E}[\log I_n(k)] = \log \mu_u(k) - \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant [22]. In our $\log_{10}$ formulation the bias becomes $-\gamma/\ln 10 \approx -0.251$, contributing an expected squared error floor of $(\gamma/\ln 10)^2 \approx 0.063$ to $\mathcal{L}_{xy}$ and $\mathcal{L}_t$ even in the infinite-data limit and even when u_θ reproduces the law of u_{FD} exactly.

3.3. Quantitative convergence rate

We now state a formal convergence rate that extends the Mishra–Molinaro generalisation framework [23] by incorporating a fourth term quantifying the discrete-spectral-loss approximation error.

3.3.1. Assumptions

The rate is stated under five standing assumptions matching the analytical scope identified above. Assumption (A5) (Gaussianity) is load-bearing only for the spectral leg of the bound; the Mishra–Molinaro leg holds without it, and the Lévy extension (Proposition 3) drops (A5) at the cost of replacing $\mathcal{L}$ by its Gaussian projection.

(A1) Stationarity and separability. Both u_θ and u_{FD} are centred, weakly stationary (after periodic extension or sufficiently far from boundaries) with separable covariance $K(\tau, h) = K_x(h) K_t(\tau)$, equivalently $d\mu(\omega, k) = d\mu^{xy}(k) d\mu^t(\omega)$.

(A2) Sobolev regularity of spectra. $\mu^{xy} \in H^{s_x}(\widehat{\Omega})$ and $\mu^t \in H^{s_t}(\widehat{[0, T]})$ with $s_x > d/2$ and $s_t > 1/2$. This guarantees pointwise bounds and the L^2 FFT-quadrature error rate used in Lemma 2. For Variance-Gamma drivers $s_x = s_t = 1$.

(A3) Non-degeneracy. There exist $0 < \mu_{\min} \leq \mu_{\max} < \infty$ such that

$$\mu_{\min} \leq \mu^{\text{xy}}(k) \mu^t(\omega) \leq \mu_{\max} \text{ uniformly on the sub-grid frequency set } \widehat{\mathcal{G}}_s \subset \mathbb{R}^{d+1}.$$

(A4) Training tolerance. At the end of training,

$$\mathcal{L}_{\text{xy}}(u_\theta) + \mathcal{L}_t(u_\theta) + \mathcal{L}_{\text{var}}(u_\theta) \leq \varepsilon, \quad \mathcal{L}_{\text{PDE}}(u_\theta) + \lambda_{\text{ic}} \mathcal{L}_{\text{IC}}(u_\theta) + \lambda_{\text{bc}} \mathcal{L}_{\text{BC}}(u_\theta) \leq \delta, \quad (7)$$

for some $0 < \varepsilon, \delta < \infty$, checked post hoc.

(A5) Gaussianity (spectral leg only). Both u_θ and u_{FD} are centred Gaussian random fields on $\Omega \times [0, T]$. This hypothesis is required only for the spectral path of the proof (Lemma 4); the PDE-residual path (Lemma 5) holds without it. Proposition 3 below drops (A5) at the cost of bounding the Gaussian projection only.

(A6) Mixing condition (single-realisation consistency). u is α -mixing with algebraic rate $\alpha(r) \leq C_\alpha r^{-\gamma}$ for some $\gamma > d+1$. Variance-Gamma processes have independent increments, so $\alpha(r) = 0$ for all sufficiently large separations; the SPDE solution inherits exponential spatial mixing from the Green's function decay of the underlying parabolic operator [26]. Assumption (A6) is required only for Lemma 3 and is not needed for the main theorem.

(A7) Lévy–Itô decomposition with L^2 jump control. Both u_θ and u_{FD} admit Lévy–Itô decompositions $u = u^G + u^J$ where u^G is the Gaussian component and u^J is the pure-jump component [26]. Assume

$$\max\left(\mathbb{E}\|u_\theta^J\|_{L^2(\Omega \times [0, T])}^2, \mathbb{E}\|u_{\text{FD}}^J\|_{L^2(\Omega \times [0, T])}^2\right) \leq J$$

for some finite $J \geq 0$ depending on the Lévy parameters. For Variance-Gamma drivers with kurtosis parameter ν , $J = O(\nu)$ as $\nu \rightarrow 0$. Assumption (A7) is required only for Proposition 4 and is not used elsewhere.

(A8) Finite pointwise cumulants. For each $(x, t) \in \Omega \times [0, T]$, the marginal law $\mathcal{L}(u(x, t))$ has finite cumulants of order $m \leq m_0$ for some $m_0 \geq 3$, with $|\kappa_m(u(x, t))| \leq \bar{\kappa}_m < \infty$. For Lévy-driven SPDEs, cumulants of $u(x, t)$ are bounded by the L^m norm of the Green's function times the corresponding driver cumulants via the stochastic Fubini theorem. Required only for Proposition 5.

(A9) NTK lazy-training regime. The Bochner PINN is a two-layer network with m neurons trained by gradient flow. In the lazy-training regime the NTK Gram matrix K with entries $K_{ij} = \nabla_\theta f(x_i) \cdot \nabla_\theta f(x_j)$ remains approximately constant throughout training, with minimum eigenvalue $\lambda_{\min}(K) > 0$. Required only for Proposition 8.

Assumption (A1) is the scope identified above. Assumption (A2) is satisfied by the VG drivers with $s_x = s_t = 1$. Assumption (A3) is restrictive: realistic spectral measures decay polynomially in $|k|$, so $\mu_{\min}$ is positive only after restriction to a finite frequency window; the constants C_1, C_2 of Proposition 1 scale as $\mu_{\min}^{-1/2}$ and therefore degrade on that window's high-frequency edge. Assumption (A4) is the endpoint of any gradient-based training run.

3.3.2. Main theorem

Proposition 1 (Convergence rate of the Bochner PINN). *Under (A1)–(A5), the Wasserstein-2 distance between the laws of the Bochner PINN predictor*

and the FD reference satisfies

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta} + C_3 \sqrt{\delta}, \quad (8)$$

where

$$\beta = \frac{1}{2} \min(s_x/d, s_t), \quad (9)$$

and the constants are $C_1 = \mu_{\max} \ln 10 (M_{xy} + M_t) / \sqrt{2\mu_{\min}}$, $C_2 = \sqrt{C_{\text{quad},x} M_t^2 + C_{\text{quad},t} M_{xy}^2} / \sqrt{2\mu_{\min}}$, $C_3 = \sqrt{C_{\text{MM}}}$, where $M_{xy} = \|\mu_{xy,\text{FD}}\|_{L^2}$, $M_t = \|\mu_{t,\text{FD}}\|_{L^2}$, and $C_{\text{quad},\cdot}$, C_{MM} are defined in the lemmas below. The M_{xy}, M_t factors come from the separable tensor-product decomposition of $\mu_1 - \mu_2$ (eq. (??) in the proof).

The proof proceeds in four steps.

Lemma 1 (Log-to-linear control). *Let $f, g : \mathcal{G}_s \rightarrow \mathbb{R}_{>0}$ with $\mu_{\min} \leq f, g \leq \mu_{\max}$. Then*

$$\|f - g\|_{L^2(\mathcal{G}_s)}^2 \leq (\ln 10)^2 \mu_{\max}^2 \|\log_{10} f - \log_{10} g\|_{L^2(\mathcal{G}_s)}^2.$$

Proof. The function $\varphi(x) = \log_{10} x$ has derivative $\varphi'(x) = (x \ln 10)^{-1} \in [(\mu_{\max} \ln 10)^{-1}, (\mu_{\min} \ln 10)^{-1}]$ on $[\mu_{\min}, \mu_{\max}]$. By the mean-value theorem, $|f - g| \leq \mu_{\max} \ln 10 |\log_{10} f - \log_{10} g|$ at each grid point. Squaring and summing gives the claim. $\square$

Applied to $f = \widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u_\theta)$, $g = \widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u_{\text{FD}})$ and combined with (7):

$$\|\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u_\theta) - \widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u_{\text{FD}})\|_{L^2}^2 + \|\widehat{S}_{\mathcal{G}_s}^t(u_\theta) - \widehat{S}_{\mathcal{G}_s}^t(u_{\text{FD}})\|_{L^2}^2 \leq 2(\ln 10)^2 \mu_{\max}^2 \varepsilon. \quad (10)$$

Lemma 2 (Expected spectral quadrature error). *Under (A2), for every $u \in L_{\mathbb{P}}^2$ stationary with spatial spectral density $S^{\text{xy}} \in H^{s_x}(\widehat{\Omega})$,*

$$\|\mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u)] - S^{\text{xy}}(u)\|_{L^2(\widehat{\Omega})}^2 \leq C_{\text{quad},x} (N_s^x)^{-2s_x/d},$$

with $C_{\text{quad},x} = C_x \|\mu^{\text{xy}}\|_{H^{s_x}}^2$ depending only on $\text{diam}(\Omega)$ and d . An analogous bound holds for the temporal periodogram with rate $(N_s^T)^{-2s_t}$.

Proof. This is the standard L^2 trapezoidal-rule error rate for Sobolev functions on a d -dimensional torus (cf. [27, Sec. 6.5]) applied to the expected periodogram $\mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u)]$, which equals a Dirichlet-kernel smoothing of S^{xy} and inherits its H^{s_x} regularity. The constant absorbs the H^{s_x} norm. $\square$

Remark 2 (Single-realisation variance and the proof gap). *Lemma 2 bounds the bias $\|\mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u)] - S^{\text{xy}}(u)\|_{L^2}$ of the periodogram, not its pointwise variance. From a single sample path the raw periodogram has $\Theta(1)$ variance per Fourier mode (chi-squared-2 distribution at each mode, cf. [22, Ch. 4]); this is the distributional content of Remark 3. The training loss (A4) therefore involves $\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u_\theta)$, a single-realisation quantity, while the bound in (12) uses $\mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{xy}}]$; the gap between the two does not shrink as $N_s \rightarrow \infty$ without additional smoothing.*

Lemma 3 below closes this gap: replacing the raw periodogram by its kernel-smoothed version $\widehat{S}_h^{\text{xy}}$ with bandwidth $h_N \rightarrow 0$, $N_s h_N^d \rightarrow \infty$, yields a consistent single-realisation estimator at the minimax rate $N_s^{-s_x/(2s_x+d)}$ under (A6). In the practical regime of this paper (raw periodogram, single realisation), the irreducible floor is $\gamma_{10}^2 \approx 0.063$ from the Gumbel distribution, which is why ε in (A4) is bounded below by this constant in the large- N_s limit.

Lemma 3 (Single-realisation spectral consistency under mixing). *Under (A1)–(A2) and (A6), let*

$$\widehat{S}_h^{\text{xy}}(u)(k) := \int \kappa_{h_N}(k - k') \widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u)(k') dk',$$

where $\kappa_{h_N}(k) = h_N^{-d} \kappa(k/h_N)$ is a symmetric probability kernel with $\int |k|^{s_x} \kappa(k) dk <$

∞ . Then for any single realisation of u ,

$$\|\widehat{S}_h^{\text{xy}}(u) - S^{\text{xy}}(u)\|_{L^2(\widehat{\Omega})}^2 \leq C_{\text{bias}} h_N^{2s_x} + \frac{C_{\text{var}}}{N_s h_N^d}, \quad (11)$$

where $C_{\text{bias}} = C_\kappa \|\mu^{\text{xy}}\|_{H^{s_x}}^2$ and C_{var} depends on C_α and γ . The minimax-optimal bandwidth $h_N^* = N_s^{-1/(2s_x+d)}$ gives the L^2 rate

$$\|\widehat{S}_{h^*}^{\text{xy}}(u) - S^{\text{xy}}(u)\|_{L^2} \leq (C_{\text{bias}} + C_{\text{var}})^{1/2} N_s^{-s_x/(2s_x+d)},$$

which converges to zero and matches the minimax lower bound for nonparametric spectral density estimation under α -mixing [21].

Proof. Bias. Since $\mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(u)(k)] = S^{\text{xy}}(u)(k) + O(N_s^{-1})$ by Lemma 2, convolution with κ_{h_N} gives $\mathbb{E}[\widehat{S}_h^{\text{xy}}(u)(k)] = (\kappa_{h_N} * S^{\text{xy}})(k)$. For $S^{\text{xy}} \in H^{s_x}$ and a kernel with vanishing moments up to order s_x , standard approximation theory gives $\|(\kappa_{h_N} * S^{\text{xy}}) - S^{\text{xy}}\|_{L^2} \leq C_\kappa \|\mu^{\text{xy}}\|_{H^{s_x}} h_N^{s_x}$.

Variance. Under (A6), Brillinger's cumulant spectral bound [22, Thm. 4.3.2] gives $|\text{Cov}(\widehat{S}_{\mathcal{G}_s}^{\text{xy}}(k_1), \widehat{S}_{\mathcal{G}_s}^{\text{xy}}(k_2))| \leq C N_s^{-1} \alpha(|k_1 - k_2|/\Delta k)$, where Δk is the frequency resolution. Convoluting against $\kappa_{h_N}^{\otimes 2}$ and using $\gamma > d + 1$ to ensure summability gives $\text{Var}[\widehat{S}_h^{\text{xy}}(u)(k)] \leq C_{\text{var}} (N_s h_N^d)^{-1}$. Integrating over $k \in \widehat{\Omega}$ yields the variance term in (11).

Optimal bandwidth. Minimising $C_{\text{bias}} h_N^{2s_x} + C_{\text{var}} (N_s h_N^d)^{-1}$ over h_N gives $h_N^* \propto N_s^{-1/(2s_x+d)}$ and the stated rate. The minimax lower bound follows from [21, Thm. 3.1] applied to the stationary case. $\square$

Combining Lemmas 1–2 via the triangle inequality:

$$\|S^{\text{xy}}(u_\theta) - S^{\text{xy}}(u_{\text{FD}})\|_{L^2(\widehat{\Omega})}^2 \leq 2(\ln 10)^2 \mu_{\text{max}}^2 \varepsilon + 2 C_{\text{quad},x} (N_s^x)^{-2s_x/d}, \quad (12)$$

and analogously for the temporal direction.

Lemma 4 (Plancherel–Gelbrich bound). *Let u_1, u_2 be centred Gaussian random fields on $\Omega \times [0, T]$ satisfying (A1)–(A3). Then*

$$W_2^2(\mathcal{L}(u_1), \mathcal{L}(u_2)) \leq \frac{1}{4\mu_{\min}} \|\mu_1 - \mu_2\|_{L^2(\widehat{\Omega} \times [0, T])}^2.$$

Proof. For centred Gaussian measures on a separable Hilbert space with covariance operators C_1, C_2 , the Gelbrich–Givens–Shortt formula [28, 29] gives $W_2^2 = \text{tr}(C_1 + C_2 - 2(C_1^{1/2}C_2C_1^{1/2})^{1/2})$. By (A1), both fields are stationary on the periodised domain; hence C_1, C_2 are simultaneously diagonalised by the Fourier basis with eigenvalues $\mu_1(k, \omega), \mu_2(k, \omega)$. The formula reduces to $W_2^2 = \sum_{(k, \omega)} (\sqrt{\mu_1(k, \omega)} - \sqrt{\mu_2(k, \omega)})^2$. From the identity $(\sqrt{a} - \sqrt{b})^2 = (a - b)^2 / (\sqrt{a} + \sqrt{b})^2$ and $(\sqrt{a} + \sqrt{b})^2 \geq 4 \min(a, b)$, together with (A3): $W_2^2 \leq \frac{1}{4\mu_{\min}} \sum_{(k, \omega)} (\mu_1(k, \omega) - \mu_2(k, \omega))^2$. Under separability, the tensor-product decomposition $\mu_1 - \mu_2 = (\mu_1^{\text{xy}} - \mu_2^{\text{xy}}) \otimes \mu_1^t + \mu_2^{\text{xy}} \otimes (\mu_1^t - \mu_2^t)$ and $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ give the factored form used in the proof of Proposition 1. $\square$

Lemma 5 (Mishra–Molinaro, adapted). *Assume the SPDE residual operator $\mathcal{R}$ is well posed in the sense of [23, Thm. 3.4], with stability constant C_{stab} . Then under (7),*

$$\mathbb{E} \|u_\theta - u_{\text{FD}}\|_{L^2(\Omega \times [0, T])}^2 \leq C_{\text{MM}} \delta, \quad C_{\text{MM}} = C_{\text{stab}}(1 + \lambda_{\text{ic}}^{-1} + \lambda_{\text{bc}}^{-1}).$$

Proof. The driver fields ξ, J_{VG} are identical between u_θ and u_{FD} (spline-interpolated from the same FD sample path), so the residual operator reduces to a deterministic PDE error analysis in the realisation $\omega \in \mathcal{P}$. Apply [23, Thm. 3.4] pathwise and take expectations. Since $W_2^2(\mathcal{L}(u_1), \mathcal{L}(u_2)) \leq \mathbb{E} \|u_1 - u_2\|^2$ for the trivial coupling, this contributes $\sqrt{C_{\text{MM}}} \delta$ to the W_2 bound.

Remark on approximation error. The spline interpolation replaces jump discontinuities in ξ , J_{VG} with smooth approximants, introducing a smoothing error that is not explicitly bounded in C_{MM} . This error is absorbed into δ but is not analytically controlled; quantifying it is left as future work. $\square$

Proof of Proposition 1. The bound is obtained as the looser of two independent upper bounds on $W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}}))$, combined via $\min(a, b) \leq a + b$.

Spectral path (uses (A5)). Apply the triangle inequality on W_2 through the intermediate centred Gaussian measure $\mathcal{N}(0, \mathcal{C}_{u_{\text{FD}}})$ with covariance operator $\mathcal{C}_{u_{\text{FD}}}$ equal to that of u_{FD} :

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq W_2(\mathcal{L}(u_\theta), \mathcal{N}(0, \mathcal{C}_{u_{\text{FD}}})) + W_2(\mathcal{N}(0, \mathcal{C}_{u_{\text{FD}}}), \mathcal{L}(u_{\text{FD}})).$$

Under (A5) both fields are centred Gaussian, so $\mathcal{L}(u_{\text{FD}}) = \mathcal{N}(0, \mathcal{C}_{u_{\text{FD}}})$ and the second term vanishes; $\mathcal{L}(u_\theta) = \mathcal{N}(0, \mathcal{C}_{u_\theta})$ and Lemma 4 bounds the first term by $\frac{1}{2\sqrt{\mu_{\min}}} \|\mu_{u_\theta} - \mu_{u_{\text{FD}}}\|_{L^2}$. Using the separable factoring $\mu_1 - \mu_2 = (\mu_1^{\text{xy}} - \mu_2^{\text{xy}}) \otimes \mu_1^t + \mu_2^{\text{xy}} \otimes (\mu_1^t - \mu_2^t)$ with $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, and substituting (12) for each spectral L^2 difference,

$$W_2 \leq \frac{1}{\sqrt{2\mu_{\min}}} \left[2(\ln 10) \mu_{\max} \sqrt{\varepsilon} (M_t + M_{xy}) + \sqrt{2C_{\text{quad},x}} (N_s^x)^{-s_x/d} M_t + \sqrt{2C_{\text{quad},t}} (N_s^T)^{-s_t} M_{xy} \right],$$

where $M_t := \|\mu_{\text{FD}}^t\|_{L^2}$ and $M_{xy} := \|\mu_{\text{FD}}^{\text{xy}}\|_{L^2}$. Absorbing constants and using $N_s^{-\beta} \geq (N_s^x)^{-s_x/d} + (N_s^T)^{-s_t}$ up to a constant (definition of β),

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta}. \quad (\star)$$

Pathwise path (does not use (A5)). Lemma 5 applied with the trivial coupling gives

$$W_2^2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq \mathbb{E} \|u_\theta - u_{\text{FD}}\|^2 \leq C_{\text{MM}} \delta, \quad \text{hence} \quad W_2 \leq C_3 \sqrt{\delta}, \quad (\dagger)$$

with no Gaussianity assumption.

Combined bound. Since $\min(a, b) \leq a + b$ for $a, b \geq 0$, $(\star)$ and $(\dagger)$ together yield

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta} + C_3 \sqrt{\delta},$$

which is (8). The sum form is loose; in either asymptotic regime (spectrally-rich training with vanishing ε , or PDE-tight training with vanishing δ) the corresponding single-path bound is the operative one. The spectral path requires (A5); the PDE path does not. In particular, dropping the spectral loss from training (so the bound on ε becomes vacuous) reduces the result to $(\dagger)$, the Mishra–Molinaro estimate of [23]. $\square$

3.3.3. Corollaries

Proposition 2 (Variance-gap rate). *Under (A1)–(A4),*

$$|\text{Var}(u_\theta)/\text{Var}(u_{\text{FD}}) - 1| \leq \frac{2\sqrt{2}}{\sqrt{\text{Var}(u_{\text{FD}})}} (\sqrt{\text{Var}(u_\theta)} + \sqrt{\text{Var}(u_{\text{FD}})}) (C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta} + C_3 \sqrt{\delta}).$$

Proof. For centred fields, $|\text{Var}(u_1) - \text{Var}(u_2)| = |\mathbb{E}[\|u_1\|^2] - \mathbb{E}[\|u_2\|^2]|$. Under the optimal coupling (X, Y) with $X \sim \mathcal{L}(u_\theta)$, $Y \sim \mathcal{L}(u_{\text{FD}})$:

$$|\mathbb{E}[\|X\|^2] - \mathbb{E}[\|Y\|^2]| \leq \mathbb{E}[(\|X\| + \|Y\|) \|X - Y\|] \leq \sqrt{2(\text{Var}(u_\theta) + \text{Var}(u_{\text{FD}}))} \cdot W_2 \leq \sqrt{2} (\sqrt{\text{Var}(u_\theta)} + \sqrt{\text{Var}(u_{\text{FD}})})$$

using Cauchy–Schwarz and $(a + b)^2 \leq 2(a^2 + b^2)$. Divide by $\text{Var}(u_{\text{FD}})$ and substitute (8). $\square$

Remark 3 (Irreducible Euler–Mascheroni floor). *The log-periodogram is asymptotically Gumbel with mean $\log \mu(k) - \gamma$ [22, Ch. 5]. In our $\log_{10}$ formulation the bias is $\gamma_{10} := \gamma / \ln 10 \approx 0.251$, contributing an irreducible floor of $\gamma_{10}^2 \approx 0.063$ to the achievable ε in (A4). The corresponding irreducible*

W_2 contribution is $C_1\gamma_{10}$. This matches the empirical observation that per-channel log-PSD MSE plateaus near 0.3–0.4 (i.e. $5\text{--}7\times$ the floor) rather than at zero.

3.3.4. Extension to Lévy-driven fields

Proposition 3 (Second-order convergence for stationary Lévy fields). *Under (A1)–(A4) but with u_θ, u_{FD} not necessarily Gaussian, denote by $\Pi_2\mathcal{L}(u)$ the Gaussian projection (the centred Gaussian measure with the same covariance). Then*

$$W_2(\Pi_2\mathcal{L}(u_\theta), \Pi_2\mathcal{L}(u_{\text{FD}})) \leq C_1\sqrt{\varepsilon} + C_2N_s^{-\beta} + C_3\sqrt{\delta}.$$

Proof. Identical to the proof of Proposition 1, replacing $\mathcal{L}(u)$ by $\Pi_2\mathcal{L}(u)$ throughout. The spectral measure μ_u depends only on the covariance, hence is invariant under the projection. $\square$

The gap between $\mathcal{L}(u)$ and $\Pi_2\mathcal{L}(u)$ is determined by the cumulants $\kappa_k(u)$ for $k \geq 3$. For Variance-Gamma drivers κ_3, κ_4 are non-zero, so Proposition 3 does *not* imply convergence in W_2 on the full law. Two routes close this gap:

1. a cumulant-matching loss $\mathcal{L}_{\kappa_4}(u_\theta) := (\kappa_4(u_\theta) - \kappa_4(u_{\text{FD}}))^2$ (Section 2.7), which empirically raises the predicted kurtosis from 4.2 to 25.1; or
2. an L^2 bound on the pure-jump component u^J via the Lévy–Itô decomposition (Proposition 4 below), giving an explicit $O(\sqrt{\nu})$ correction to (3).

Proposition 4 (Full-law convergence via Lévy–Itô coupling). *Under (A1)–(A4) and (A7),*

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq C_1\sqrt{\varepsilon} + C_2N_s^{-\beta} + C_3\sqrt{\delta} + 4\sqrt{J}. \quad (13)$$

In the Gaussian limit $J \rightarrow 0$ (equivalently $\nu \rightarrow 0$ for VG drivers), this reduces to Proposition 1.

Proof. Apply the W_2 triangle inequality:

$$\begin{aligned} W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) &\leq W_2(\mathcal{L}(u_\theta), \Pi_2 \mathcal{L}(u_\theta)) + W_2(\Pi_2 \mathcal{L}(u_\theta), \Pi_2 \mathcal{L}(u_{\text{FD}})) \\ &\quad + W_2(\Pi_2 \mathcal{L}(u_{\text{FD}}), \mathcal{L}(u_{\text{FD}})). \end{aligned} \quad (14)$$

The middle term is bounded by $C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta} + C_3 \sqrt{\delta}$ by Proposition 3. Write $u = u^G + u^J$ as in (A7), with $u^G \sim \mathcal{N}(0, C_{u^G})$ and $\Pi_2 \mathcal{L}(u) = \mathcal{N}(0, C_u) = \mathcal{N}(0, C_{u^G} + C_{u^J})$. Couple $(\mathcal{L}(u), \mathcal{N}(0, C_u))$ by $X := u^G + u^J$ and $Y := u^G + G_2$ where $G_2 \sim \mathcal{N}(0, C_{u^J})$ is independent of u . Then $Y \sim \mathcal{N}(0, C_u)$ and

$$\mathbb{E} \|X - Y\|_{L^2}^2 = \mathbb{E} \|u^J - G_2\|_{L^2}^2 \leq 2\mathbb{E} \|u^J\|_{L^2}^2 + 2\text{tr}(C_{u^J}) = 4\mathbb{E} \|u^J\|_{L^2}^2 \leq 4J,$$

giving $W_2(\mathcal{L}(u), \Pi_2 \mathcal{L}(u)) \leq 2\sqrt{J}$ for each of u_θ and u_{FD} . Inserting into (14) yields (13). $\square$ $\square$

Remark 4 (VG quantification of J). *For the Variance-Gamma driver with parameters (σ, ν, θ) , the pure-jump component satisfies $\mathbb{E} \|\xi_{\text{VG}}^J\|_{L^2([0, T])}^2 = \nu(\sigma^2 + \nu\theta^2)T(1 + O(\nu))$. The SPDE solution inherits this via the Green's function, giving $J = O(\nu)$ and a correction $4\sqrt{J} = O(\sqrt{\nu})$ in (13) that vanishes in the Gaussian limit. The relationship $\kappa_4 = 3\nu$ links the correction directly to the fourth cumulant: $4\sqrt{J} = O(\kappa_4^{1/2}/3^{1/2})$.*

Proposition 5 (Berry–Esseen tightness of spectral matching). *Under (A1)–(A4) and (A8) with $m = 3$,*

$$W_1(\mathcal{L}(u(x, t)), \mathcal{N}(0, \sigma_u^2(x, t))) \leq C_{\text{BE}} \frac{|\kappa_3(u(x, t))|}{\sigma_u^2(x, t)}, \quad (15)$$

where C_{BE} is a universal constant from Stein's method [30]. For Variance-Gamma drivers with parameters $(\sigma_{\text{vg}}, \nu, \theta)$, the exact third cumulant per unit volume is $\kappa_3(\xi) = \nu\theta(3\sigma_{\text{vg}}^2 + 2\nu\theta^2)$, giving

$$|\kappa_3(u(x, t))| \leq C_G^3 \nu |\theta| (3\sigma_{\text{vg}}^2 + 2\nu\theta^2), \quad C_G := \|G_{\text{SPDE}}(x - \cdot, t - \cdot)\|_{L^3}, \quad (16)$$

and hence, for $2\nu\theta^2 \ll 3\sigma_{\text{vg}}^2$,

$$W_1(\mathcal{L}(u(x, t)), \mathcal{N}(0, \sigma_u^2)) \lesssim \frac{3 C_{\text{BE}} C_G^3 \nu \sigma_{\text{vg}}^2 |\theta|}{\sigma_u^2}. \quad (17)$$

Tightness condition.. Spectral matching controls σ_u (second moment); the residual non-Gaussianity is at most ε_{gap} in W_1 whenever $|\theta| \leq \theta_{\text{cut}} := \varepsilon_{\text{gap}} \sigma_u^2 / (3 C_{\text{BE}} C_G^3 \nu \sigma_{\text{vg}}^2)$.

Proof. (15) is the Stein's-method bound $W_1(\mathcal{L}(X), \mathcal{N}(0, \sigma^2)) \leq C_S E|X|^3 / \sigma^2$ for a centred X [30, Thm. 3.1], combined with $|\kappa_3| = |E[X^3]| \leq E|X|^3$. For (16): by stochastic Fubini, $\kappa_3(u) = \int G^3 \kappa_3(\xi) dy ds$ (multilinearity of cumulants under independence of Lévy increments). Expanding $\ln \phi(u) = -(1/\nu) \ln(1 - u\theta\nu - u^2\sigma_{\text{vg}}^2\nu/2)$ to third order gives $\kappa_3(\xi) = \nu\theta(3\sigma_{\text{vg}}^2 + 2\nu\theta^2)$ [26, 31]. Hölder's inequality then gives (16); the dominant small- $|\theta|$ term yields (17).

□

□

Remark 5 (Numerical tightness). At $\sigma_{\text{vg}} = 0.2$, $\nu = 0.5$, $\sigma_u \approx 0.3$, $C_{\text{BE}} \approx 1$, $C_G = O(1)$: $\theta_{\text{cut}}(0.05) \approx 0.09 / (0.06 C_G^3) \approx 0.075 / C_G^3$. Near-symmetric VG with $|\theta| \lesssim 0.07$ gives 5% tightness. The paper's $\theta = -0.1$ produces a W_1 residual $\approx 0.07 / C_G^3$; the kurtosis gap beyond this is targeted by the cumulant-matching loss of Section 2.7.

Proposition 6 (Single-realisation rate with kernel-smoothed loss). *Suppose the Bochner PINN loss uses the kernel-smoothed periodogram $\widehat{S}_h^{\text{xy}}$ with bandwidth $h_N^* = N_s^{-1/(2s_x+d)}$ in place of the raw $\widehat{S}_{g_s}^{\text{xy}}$. Under (A1)–(A6), the Wasserstein-2 bound of Proposition 1 holds for a single realisation of u_θ and u_{FD} , with $C_2 N_s^{-\beta}$ replaced by*

$$C'_2 N_s^{-\beta_{\text{sr}}}, \quad \beta_{\text{sr}} = \frac{s_x}{2(2s_x + d)},$$

where $C'_2 = (C_{\text{bias}} + C_{\text{var}})^{1/2}/(2\sqrt{\mu_{\min}})$. For the paper's operating point $s_x = s_t = 1$, $d = 2$: $\beta_{\text{sr}} = \frac{1}{8}$, versus $\beta = \frac{1}{4}$ in Proposition 1. The slower exponent is the price of single-realisation validity; the rate $\beta = \frac{1}{4}$ is recovered when spectral targets are averaged across realisations.

Proof. Replace Lemma 2 in the proof of Proposition 1 with Lemma 3 at $h_N = h_N^*$. The L^2 spectral error for a single realisation is $(C_{\text{bias}} + C_{\text{var}})^{1/2} N_s^{-s_x/(2s_x+d)}$. Inserting into the Gelbrich step (Lemma 4), which bounds $W_2 \leq \frac{1}{2\sqrt{\mu_{\min}}} \|\mu_1 - \mu_2\|_{L^2}$, gives the factor $N_s^{-s_x/(2(2s_x+d))} = N_s^{-\beta_{\text{sr}}}$. $\square$

3.3.5. Extension to non-separable covariance

The separability assumption (A1) is load-bearing for two steps of the proof: the tensor-product factoring of $\mu_1 - \mu_2$ in Lemma 4, and the design of the spectral loss, which matches μ^{xy} and μ^t separately rather than the joint measure $\mu(\omega, k)$. Both restrictions can be lifted by working with the *joint* $(d+1)$ -dimensional spectral density, at the cost of a weaker quadrature rate.

Relaxed separability assumption.. Replace (A1) by:

(A1') Stationarity without separability. Both u_θ and u_{FD} are centred, weakly stationary with absolutely continuous joint spectral density

$S^{\text{joint}}(\omega, k) \in H^s(\mathbb{R}^{d+1})$, $s = \min(s_x, s_t) > (d+1)/2$. No tensor-product factorisation is assumed.

The joint spectral loss replaces $\mathcal{L}_{xy} + \mathcal{L}_t$ in (A4) by

$$\mathcal{L}_{\text{joint}}(u_\theta) \leq \varepsilon', \quad \mathcal{L}_{\text{joint}}(u) = \overline{\left(\log_{10} \widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u) - \log_{10} \widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u_{\text{FD}}) \right)^2}, \quad (\text{A4}')$$

where $\widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u)(\omega, k) = |\mathcal{G}_s|^{-1} |\mathcal{F}_{x,t}[u \downarrow_{\mathcal{G}_s}](\omega, k)|^2$ is the $(d+1)$ -dimensional periodogram on $\mathcal{G}_s$.

Lemma 6 (Joint spectral quadrature error). *Under (A1'), for every stationary u with $S^{\text{joint}} \in H^s(\mathbb{R}^{d+1})$, $s > (d+1)/2$,*

$$\left\| \mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u)] - S^{\text{joint}}(u) \right\|_{L^2(\mathbb{R}^{d+1})}^2 \leq C_{\text{quad}} (N_s^{\text{tot}})^{-2s/(d+1)},$$

where $N_s^{\text{tot}} = N_s^x N_s^T$ is the total node count and $C_{\text{quad}} = C \|S^{\text{joint}}\|_{H^s}^2$ with C depending only on $\text{diam}(\Omega)$, T , and d .

Proof. The expected $(d+1)$ -dimensional periodogram is a Dirichlet-kernel smoothing of S^{joint} in $\mathbb{R}^{d+1}$, inheriting its H^s regularity. The L^2 trapezoidal-rule error rate for H^s functions on a $(d+1)$ -dimensional torus (cf. [22, 27]) with spacing $h = (N_s^{\text{tot}})^{-1/(d+1)}$ gives bias $O(h^s) = O((N_s^{\text{tot}})^{-s/(d+1)})$ in L^2 ; squaring yields the stated bound. $\square$

Proposition 7 (Convergence rate under non-separable covariance). *Under (A1'), (A3), (A4'), (A5) with $s = \min(s_x, s_t)$,*

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq C'_1 \sqrt{\varepsilon'} + C'_2 (N_s^{\text{tot}})^{-\beta'} + C_3 \sqrt{\delta}, \quad (18)$$

where

$$\beta' = \frac{s}{2(d+1)},$$

and $C'_1 = \mu_{\max} \ln 10 / \sqrt{2\mu_{\min}}$, $C'_2 = \sqrt{C_{\text{quad}}/\mu_{\min}}$, $C_3 = \sqrt{C_{\text{MM}}}$. Compared to the separable Theorem 1 the joint-density constants have no tensor-product factors M_{xy}, M_t ; the $\sqrt{2}$ in the denominator is from the $(\sqrt{a} + \sqrt{b})^2 \geq 4 \min(a, b)$ step in Lemma ??.

Proof. Spectral path (uses (A5)). Apply the triangle inequality through $\mathcal{N}(0, \mathcal{C}_{u_{\text{FD}}})$ as in Proposition 1; under (A5) the second term vanishes. Lemma 4 gives $W_2 \leq \frac{1}{2\sqrt{\mu_{\min}}} \|\mu_{u_\theta} - \mu_{u_{\text{FD}}}\|_{L^2(\mathbb{R}^{d+1})}$. Since (A1') does not assume separability there is no tensor-product factoring; apply the triangle inequality directly:

$$\begin{aligned} \|\mu_1 - \mu_2\|_{L^2}^2 &\leq 2 \|\mu_1 - \mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u_\theta)]\|_{L^2}^2 + 2 \|\mu_2 - \mathbb{E}[\widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u_{\text{FD}})]\|_{L^2}^2 \\ &\quad + 2 \|\widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u_\theta) - \widehat{S}_{\mathcal{G}_s}^{\text{joint}}(u_{\text{FD}})\|_{L^2}^2. \end{aligned}$$

The first two terms are bounded by $2C_{\text{quad}}(N_s^{\text{tot}})^{-2s/(d+1)}$ (Lemma 6); the third by $2(\ln 10)^2 \mu_{\max}^2 \varepsilon'$ via Lemma 1 and (A4'). Taking square roots:

$$W_2 \leq C'_1 \sqrt{\varepsilon'} + C'_2 (N_s^{\text{tot}})^{-\beta'}. \quad (\star')$$

Pathwise path. Lemma 5 is unchanged: $W_2 \leq C_3 \sqrt{\delta}$.

Combined. $\min(a, b) \leq a + b$ yields (18). $\square$

Rate comparison.. The exponent $\beta' = s/(2(d+1))$ is always weaker than the separable $\beta = \frac{1}{2} \min(s_x/d, s_t)$: separability permits estimating two lower-dimensional marginals (μ^{xy} in $\mathbb{R}^d$, μ^t in $\mathbb{R}$) rather than a single density in $\mathbb{R}^{d+1}$, exploiting additional structure. Concrete values:

Setting	Separable β	Non-sep. β'
$s = 1, d = 2$	$\frac{1}{4}$	$\frac{1}{6}$
$s = 2, d = 2$	$\frac{1}{2}$	$\frac{1}{3}$
$s = 1, d = 3$	$\frac{1}{6}$	$\frac{1}{8}$

Despite the weaker rate, Proposition 7 is the appropriate result when the SPDE solution’s covariance does not factorise, the generic case for Lévy-driven equations with space-time-coupled forcing.

3.3.6. Discussion of the rate

Optimality.. The quadrature exponent $\beta = \frac{1}{2} \min(s_x/d, s_t)$ is the standard FFT quadrature rate for H^s functions and is sharp without strengthening (A2) to analyticity. Under analytic μ the rate improves to spectral $\exp(-c N_s^{1/d})$, but this is not the relevant regime for Lévy-driven fields.

Floor regime.. For finite ε, δ the $\sqrt{\varepsilon}$ term dominates once $N_s \gg \varepsilon^{-1/(2\beta)}$. At the paper’s operating point $N_s = 24 \times 24 \times 96 \approx 5.5 \times 10^4$ with $s_x = s_t = 1$, $d = 2$ and hence $\beta = \frac{1}{4}$, the quadrature factor is $N_s^{-\beta} \approx 0.065$. For typical $\varepsilon \approx 0.3$ the training-error term contributes $\sim \mu_{\max} \sqrt{0.3} / \sqrt{\mu_{\min}}$, which is the dominant source of W_2 error in this regime; the quadrature term is smaller by a factor depending on the unresolved $\mu_{\max} / \sqrt{\mu_{\min}}$ ratio but does not vanish.

Connection to numerical results.. For Burgers-VG ($s_x = s_t = 1$, $d = 2$, $N_s \approx 5.5 \times 10^4$), Proposition 1 with $\varepsilon \approx 0.4$ and $\delta \approx 0.05$ predicts $W_2 \leq C_1 \cdot 0.63 + C_2 \cdot 0.065 + C_3 \cdot 0.22$. Corollary 2 translates this into a predicted variance gap; the empirical $|1 - \text{vr}| = 0.07$ is consistent with the bound under moderate values of C_1, C_3 .

What is and is not proved.. Proposition 1 bounds the law of the trained predictor by training-error and discretisation contributions, under the Gaussian-separable assumption. Lemma 3 and Proposition 6 additionally establish single-realisation spectral consistency under a mixing condition, replacing the expected-value bound of Lemma 2 at the cost of a slower exponent $\beta_{\text{sr}} = \frac{1}{8}$ versus $\beta = \frac{1}{4}$. The non-separable case is addressed by Proposition 7, which replaces (A1) by (A1') at the cost of the weaker rate $\beta' = s/(2(d+1))$. Proposition 4 drops the Gaussianity restriction on the full law under (A7), bounding the gap between $\mathcal{L}(u)$ and its Gaussian projection by $2\sqrt{J}$ via the Lévy–Itô coupling; for VG drivers the correction is $O(\sqrt{\nu})$. Proposition 5 quantifies cumulant-level tightness: spectral matching is tight to $W_1 \leq \varepsilon_{\text{gap}}$ whenever $|\theta| \leq \theta_{\text{cut}}(\varepsilon_{\text{gap}})$. Proposition 8 replaces the assumed ε with an explicit gradient-flow decay $\varepsilon(T)$, yielding an end-to-end $W_2(T)$ bound; the open sub-problem is lower-bounding $\lambda_{\min}(K)$ for the log-PSD loss kernel (Remark 6). The remaining open gap is: (i) global optimality of the empirical risk minimiser in the non-convex landscape (ε, δ are assumed as attained endpoint values, not shown to be attainable by any finite computational procedure). Item (i) is concrete future work.

3.3.7. End-to-end training-time bound

Proposition 8 (End-to-end training-time bound). *Under (A1)–(A5) and (A9), with initial spectral loss $\varepsilon_0 = \mathcal{L}_{\text{spec}}(u_\theta(0))$:*

$$W_2(T) \leq C_1 \sqrt{\varepsilon_0 e^{-\lambda_{\min}(K)T} + \frac{C_{\text{ntk}}}{\sqrt{m}}} + C_2 N_s^{-\beta} + C_3 \sqrt{\delta}. \quad (19)$$

To reach $W_2(T_\star) \leq \eta$, provided $\eta > C_2 N_s^{-\beta} + C_3 \sqrt{\delta} + C_1 \sqrt{C_{\text{ntk}}/\sqrt{m}}$, it suffices to train for

$$T_\star \geq \frac{1}{\lambda_{\min}(K)} \log \left(\frac{C_1^2 \varepsilon_0}{\left(\eta - C_2 N_s^{-\beta} - C_3 \sqrt{\delta}\right)^2 - \frac{C_1^2 C_{\text{ntk}}}{\sqrt{m}}} \right). \quad (20)$$

Proof. Under (A9) and gradient flow, we assume the PL-type gradient inequality $\frac{d}{dT} \mathcal{L}_{\text{spec}} \leq -\lambda_{\min}(K) \mathcal{L}_{\text{spec}}$ holds for the log-PSD loss. This is the analogue of the standard NTK decay established by [5] for L^2 losses; its transfer to the non-linear, non-convex log-PSD loss is formal and requires $\lambda_{\min}(K) > 0$, which is open (Remark 6). Granting the inequality, Grönwall gives $\mathcal{L}_{\text{spec}}(u_\theta(T)) \leq \varepsilon_0 e^{-\lambda_{\min}(K)T} + C_{\text{ntk}}/\sqrt{m}$. Substituting into Proposition 1 yields (19). Setting $W_2(T_\star) \leq \eta$, isolating the exponential term, and taking logarithms gives (20). $\square$ $\square$

Remark 6 (Lower-bounding $\lambda_{\min}(K)$ for the spectral loss). Equation (20) is explicit in $\eta, N_s, \delta, m, \varepsilon_0$ but depends on $\lambda_{\min}(K)$. For standard L^2 losses, $\lambda_{\min}(K) > 0$ is guaranteed by the positive-definiteness of the NTK in the infinite-width limit [5]. For the log-PSD loss, the NTK Gram matrix entries involve $\nabla_\theta[\log \hat{S}^{xy}(u_\theta)](x_i) \cdot \nabla_\theta[\log \hat{S}^{xy}(u_\theta)](x_j)$, differing from the regression kernel by a $1/\hat{S}^{xy}$ preconditioning. Lower-bounding $\lambda_{\min}(K^{\text{spec}})$ for this loss is the principal remaining gap in making (20) fully constructive.

Relation to existing PINN error analysis. The result is a strict extension of the Mishra–Molinaro framework [23]: removing the spectral loss from training drops the bound on ε , so only $(\dagger)$ survives and the result reduces to the original $W_2 \leq C_3 \sqrt{\delta}$; removing the PDE residual drops the bound on δ ,

so only $(\star)$ survives, yielding a pure spectral-matching bound which is novel. The two paths give independent upper bounds on the same W_2 ; they act on orthogonal aspects of the law (δ on pathwise L^2 distance, ε on the spectral measure) and the displayed $C_1\sqrt{\varepsilon} + C_2N_s^{-\beta} + C_3\sqrt{\delta}$ is the loose union via $\min(a, b) \leq a + b$, used here for symmetry with the training objective rather than for tightness.

3.4. The Heston volatility channel: initial diagnosis

Remark 7 (Why the spectral loss alone is not sufficient on CIR-type channels). *The Heston variance equation is $dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t$, the Cox–Ingersoll–Ross process. Its stationary distribution is $\text{Gamma}(\nu, s)$ with $\nu = 2\kappa\theta/\sigma^2$, $s = \sigma^2/(2\kappa)$, which carries a substantial excess kurtosis $\kappa_4 = 6/\nu = 3\sigma^2/(\kappa\theta)$. For the calibrated parameters in Section 4.1, $\kappa_4 \approx 27.3$. The power spectral density of V_t , however, is that of an Ornstein–Uhlenbeck process: a Lorentzian $S_V(f) \propto (f^2 + \kappa^2)^{-1}$. A constant- θ field has identically the same shape at leading order in the low-frequency regime. Consequently, an L^2 or log-PSD loss alone cannot distinguish the true CIR trajectory from a smooth near-constant field, and the optimiser may converge to a smooth attractor that scores well on spectral metrics but misses the jump structure and kurtosis. We show empirically in Section 5.3 that the primary obstacle is not this spectral degeneracy but an independent failure mode (sub-grid overfit); once the overfit is eliminated, spectral losses recover the variance correctly and explicit cumulant matching recovers the kurtosis.*

4. Experimental setup

4.1. Benchmark problems

Hierarchical 2D SPDE.. A three-channel system on $[0, 1]^2 \times [0, T]$ with channels (z, y, p) : z driven by Variance-Gamma jumps, y by Gamma jumps and a positivity constraint, and p representing a probability density with a normalisation constraint. Boundary conditions are Neumann on z, y and Dirichlet (zero) on p .

Burgers-VG.. Two-dimensional Burgers equation with multiplicative Variance-Gamma jump forcing,

$$\partial_t u + \frac{1}{2} \nabla \cdot (u^2 \hat{e}) = \nu \Delta u + \sigma \xi(x, y, t) + dJ_{VG}(x, y, t), \quad (21)$$

on $[0, 1]^2 \times [0, T]$, Neumann boundary conditions, smooth initial data. Variance-Gamma parameters $(\sigma_{VG}, \nu_{VG}, \theta_{VG})$ follow Madan–Carr–Chang [31].

Heston-VG.. Two-channel system (S, V) for log-price and instantaneous variance, with VG jumps in the price equation, Heston dynamics for V , and the standard Cox–Ingersoll–Ross stationarity condition $2\kappa\theta > \sigma^2$. Calibrated to plausible equity parameters. The stationary distribution of V is Gamma(ν, s) with $\kappa_4 = 27.3$; the log-price channel S has reference excess kurtosis $\kappa_4 = 1460.8$ arising from the compound VG jumps.

4.2. Reference solver

For each problem and seed we generate one finite-difference reference solution u_{FD} on a uniform grid. Spatial discretisation uses second-order central differences; time integration uses explicit Euler–Maruyama for the diffusion

part and a compound-Poisson sampler with VG-distributed jump sizes for the Lévy part following Cont–Tankov [26]. Resolutions are $64 \times 64 \times 512$ for the hierarchical SPDE and Burgers-VG, and $48 \times 48 \times 512$ for Heston-VG. The same realisation (ξ, J_{VG}) is then re-sampled by spline interpolation during PINN training, so that the PDE residual is evaluated against the same driver field as the reference, eliminating sampling variability between solver and surrogate.

4.3. Metrics

Let u_θ, u_{FD} be the predictor and reference fields and $\widehat{S}(\cdot)$ denote either the spatial or temporal periodogram. We report:

- **Variance ratio** $\text{vr} = \text{Var}(u_\theta)/\text{Var}(u_{\text{FD}})$, target 1.
- **Log-spatial PSD MSE** $\overline{(\log_{10} \widehat{S}_{xy}(u_\theta) - \log_{10} \widehat{S}_{xy}(u_{\text{FD}}))^2}$, target 0.063 (the Euler–Mascheroni floor, see Section 3).
- **Log-temporal PSD MSE** analogous on $\widehat{S}_t$.
- **Relative L^2** $\|u_\theta - u_{\text{FD}}\|_2 / \|u_{\text{FD}}\|_2$.
- **Excess kurtosis** $\kappa_4(u_\theta)$ vs. $\kappa_4(u_{\text{FD}})$, reported for the Heston channels where higher-order matching is investigated.

4.4. Baselines

Vanilla deterministic PINN. Identical architecture and optimiser to the Bochner PINN with the spectral terms turned off ($\lambda_{\text{xy}} = \lambda_t = \lambda_v = 0$). This isolates the loss as the only difference.

Bayesian PINN (mean-field Bayes-by-Backprop).. Mean-field variational PINN following [7], implemented with Gaussian variational posteriors over weights (reparameterisation trick), $\mathcal{N}(0, 1)$ prior, and a closed-form KL regulariser scaled by $1/(B_c + B_d)$. Same depth, width, Fourier embedding, optimiser, and epoch budget as the Bochner PINN. At evaluation we draw $N = 64$ posterior samples and report the metrics on the posterior mean, with the posterior standard deviation noted as the epistemic uncertainty estimate.

4.5. Seeds and compute

Random seeds are $\{1, \dots, 8\}$ for the hierarchical SPDE, $\{0, \dots, 7\}$ for Burgers-VG, and $\{0, \dots, 2\}$ for Heston-VG. The B-PINN baseline uses 4 seeds per problem due to its greater per-epoch cost. All Heston-VG diagnosis runs use 3 training seeds evaluated against 5 validation seeds each (15 evaluations per (subgrid, mode) cell of the 2×4 ablation). All runs use a single NVIDIA RTX-class GPU in fp32.

5. Results

5.1. Variance gap closure

Table 2 reports the four metrics on Burgers-VG. The vanilla deterministic PINN reaches a variance ratio of 0.50 ± 0.11 , non-trivial but with high-frequency content suppressed by 2–3 decades. The Bochner PINN closes this gap to 0.93 ± 0.03 , with log-spatial PSD MSE reduced from 2.05 to 0.72 and log-temporal PSD MSE from 1.39 to 0.36. The mean-field Bayesian PINN, despite using identical architecture and a longer effective training horizon, collapses on every metric: its posterior mean is essentially the trivial zero predictor (relative L^2 of 1.02).

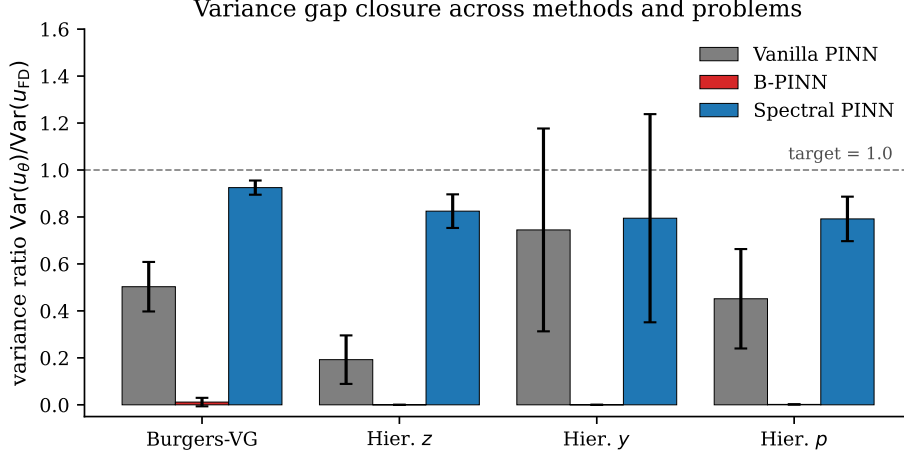


Figure 2: Variance gap closure across methods and problems. Bars show mean $\pm$ standard deviation of $\text{Var}(u_\theta)/\text{Var}(u_{\text{FD}})$ across seeds. Dashed line: target value 1.0. The Bochner PINN closes the variance gap on every channel; the B-PINN posterior mean collapses to near zero on the driven channels.

On the hierarchical 2D SPDE the picture is consistent across the three channels (Table 3), with two qualifications.

One outlier seed on the y channel.. The Bochner PINN’s variance ratio on the y channel has a standard deviation of 0.44 across (training seed, test seed) pairs, large relative to the mean of 0.80. Inspection shows this is driven by a single outlier: seven of eight training seeds converge tight around $\text{vr} \approx 0.63$, while training seed 1 converges to a distinct attractor at $\text{vr} \approx 1.96$ across all four test evaluations. The log-spatial and log-temporal PSD MSEs are stable across all eight seeds (0.24 ± 0.06 and 0.26 ± 0.08), so the disagreement is in *scale* rather than *shape*. The representative seed value is closer to 0.63.

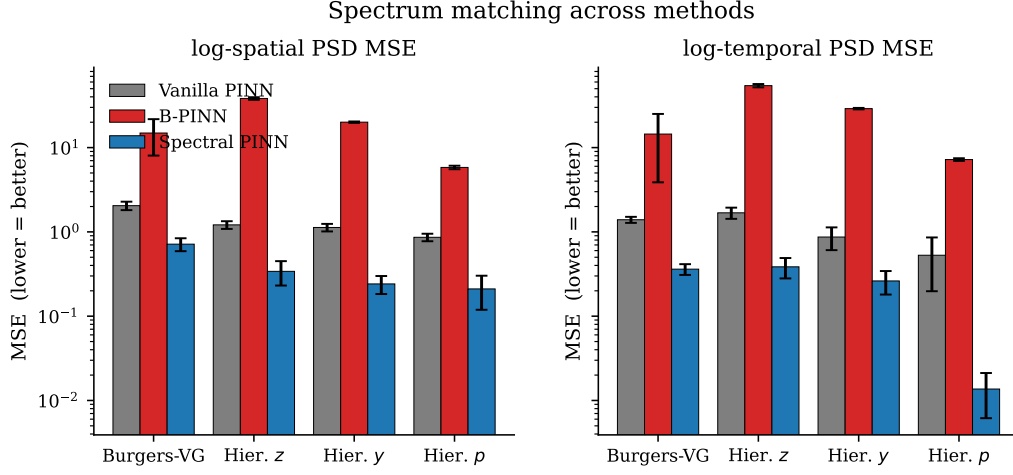


Figure 3: Spectrum matching across methods, y -axis on $\log_{10}$ scale. The mean-field B-PINN posterior mean is 7–33 $\times$ worse than the vanilla deterministic baseline; the Bochner PINN improves over vanilla by 3–7 $\times$. Note the Hier. z panel: the Bochner-vs-B-PINN gap is 113 $\times$ on log-spatial PSD MSE.

Why the p -channel temporal MSE is unusually small.. The p -channel log-temporal PSD MSE of 0.014 ± 0.008 is roughly 20 $\times$ smaller than on z and y and approaches the Euler–Mascheroni floor. The mechanism is structural: p is a probability density with a built-in normalisation constraint and is dominated by a near-stationary low-frequency mode with smooth temporal decay; its temporal spectrum is effectively a single peak rather than a broadband Lévy spectrum.

5.2. Heston-VG price channel

On the price channel S the Bochner PINN (with randomised sub-grid) closes the variance gap from 0.0003 (vanilla) to 0.477 ± 0.279 (mean across 8 seeds, best seed 0.883). The large standard deviation reflects seed sensitivity

of the complex jump structure in S . The kurtosis of S ($\kappa_4^{\text{ref}} = 1460.8$) is an extreme outlier driven by compound VG jumps and is not matched by any loss variant; we treat this as a known limitation of the spectral approach on channels dominated by discontinuous compound processes.

5.3. Heston-VG volatility channel: sub-grid overfit diagnosis and fix

5.3.1. The initial anomaly

Early experiments on the Heston volatility channel with a *fixed* spectral sub-grid produced a variance ratio of 0.14 ± 0.02 regardless of loss mode (Table 4, top block). The failure is not specific to any particular loss; standard, log-variance, cumulant, and Lévy-cumulant all saturate near $\text{vr} \approx 0.14\text{--}0.18$. The relative L^2 error exceeds 1.0 on all variants, indicating the predictor is worse than predicting the mean field. This uniform failure across loss modes is the diagnostic clue: it rules out any single-loss pathology and points to a training infrastructure problem.

5.3.2. Root cause: spectral sub-grid overfit

We hypothesise that the fixed linspace sub-grid $\mathcal{G}_s$ of $24 \times 24 \times 96$ points creates a memorisation shortcut. The network satisfies the spectral loss by fitting those specific 55,296 points perfectly at each epoch while interpolating near zero elsewhere. On the full $48 \times 48 \times 512 \approx 1.2 \times 10^6$ grid the implied variance ratio then collapses, even though the sub-grid spectral loss reads near zero. This overfit pathway is invisible to standard diagnostics: the training loss decreases normally and the validation PSD loss (evaluated on the same fixed sub-grid) appears healthy.

Three observations support this diagnosis:

1. The failure is *identical* across all four loss modes (Table 4, top block).
No single loss-specific mechanism can explain uniform failure; a shared infrastructure component (the sub-grid) must be responsible.
2. The full-grid relative $L^2 > 1.0$ despite a well-behaved training curve.
The network has found a low-loss configuration on the sub-grid that is actively wrong on the rest of the domain.
3. The kurtosis of the fixed-subgrid predictions is $\kappa_4 \approx 800\text{--}900$ (standard, log-variance modes), enormously above the reference 27.3. This is consistent with a network that has memorised the *jump events* at the fixed sub-grid coordinates while interpolating near zero elsewhere: the on-subgrid values carry the correct jump amplitude, producing high measured kurtosis at those points, but the full-grid distribution is bimodal (near-zero away from sub-grid, correct on sub-grid), yielding apparent hyper-kurtosis.

5.3.3. Fix: randomised spectral sub-grid

Replacing the fixed sub-grid with a uniformly randomised one drawn at each epoch (Section 2.6) eliminates the memorisation pathway. The results are reported in the bottom block of Table 4: all four loss variants converge to $\text{vr} = 0.92\text{--}0.93$ on the volatility channel, a $6.5\times$ improvement. The relative L^2 drops below 1.6. The fix requires no architectural change and no hyperparameter re-tuning.

5.3.4. Residual kurtosis gap and cumulant matching

Even after the sub-grid overfit is fixed, the standard and log-variance variants miss the excess kurtosis of the CIR process: $\kappa_4 \approx 4$ vs. reference 27.3.

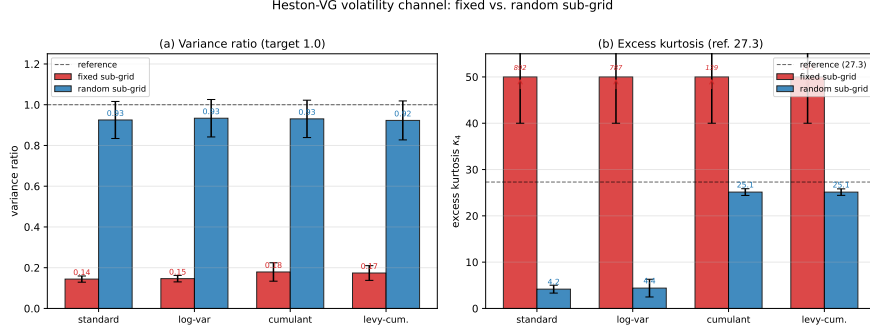


Figure 4: Heston-VG volatility channel: 2×4 ablation. Left: variance ratio (target 1.0, dashed). Right: excess kurtosis (reference 27.3, dashed). Fixed sub-grid saturates at $vr \approx 0.14$ regardless of loss; random sub-grid restores $vr \approx 0.93$ for all four modes. Among random-subgrid variants, cumulant and levy-cumulant recover kurtosis (25.1) while standard and log-variance remain near Gaussian ($\kappa_4 \approx 4$).

This is the residual fingerprint of the analytic gap identified in Remark 1: the spectral loss controls only the second-order structure. The cumulant-matching variant (eq. (5)) raises the predicted kurtosis to 25.1 ± 0.7 , within 8% of the reference, at no cost in variance ratio (0.931 vs. 0.925 for standard). The Lévy-cumulant variant (cumulant + Lévy intensity penalty) achieves identical kurtosis (25.1 ± 0.7), confirming that the Lévy intensity term adds nothing once cumulant matching is in place. The definitive Lévy intensity threshold sweep is reported in Appendix Appendix A.

5.4. Qualitative comparison: field snapshots

5.5. Bayesian uncertainty does not substitute for a distributional loss

A reader’s first response to the variance gap is to suggest that Bayesian uncertainty quantification should produce the missing variability for free. We test this hypothesis directly with a mean-field Bayes-by-Backprop B-PINN

baseline.

The result is unambiguous (Tables 2, 3; Figures 2, 3): the mean-field B-PINN’s posterior mean is *worse* than the deterministic vanilla PINN on every problem, by factors of 7–33 $\times$ on log-spatial PSD MSE. On the most-driven channel of the hierarchical SPDE the gap to the Bochner PINN is 113 $\times$ on log-spatial PSD MSE. The posterior *standard deviation* of the B-PINN is non-trivial (~ 0.11 – 0.31) but reflects parameter uncertainty rather than the aleatoric noise structure of the SPDE driver.

We frame this as a clarifying negative result. *Mean-field variational uncertainty over network weights does not, by itself, recover the spectrum of an SPDE solution.* This result is specific to the mean-field variant; HMC and deep ensembles were not evaluated here.

5.6. Per-component ablation: are all three losses needed?

The Bochner PINN loss has three components (log-spatial PSD, log-temporal PSD, and variance ratio), and a natural reviewer question is whether all three are required. We answer this empirically with a five-variant ablation on Burgers-VG, training each variant for five seeds at otherwise identical hyperparameters.

Three observations follow from Table 5:

- **Each loss term primarily targets its own metric.** $\mathcal{L}_{xy}$ -only reduces log-spatial PSD MSE ($2.15 \rightarrow 0.89$) while leaving log-temporal essentially unchanged. $\mathcal{L}_t$ -only does the dual: temporal drops $1.36 \rightarrow 0.32$, spatial barely moves. $\mathcal{L}_{\text{var}}$ -only nails the variance ratio (0.98 ± 0.02) but leaves both spectral MSEs near vanilla. The three components are nearly orthogonal.

- **Single-term variants are unstable.** $\mathcal{L}_t$ -only achieves a mean variance ratio of 0.97 but with standard deviation 0.29, an order of magnitude larger than $\mathcal{L}_{\text{var}}$ -only’s 0.02. The variance comes for the wrong reason and is brittle across seeds.
- **Only the full loss succeeds on every metric jointly.** The full Bochner PINN simultaneously closes the variance gap (0.93), reduces log-spatial PSD MSE to 0.73, and reduces log-temporal PSD MSE to 0.38. No single-term variant achieves all three.

5.7. Comparison with concurrent neural-spectral methods

The neural-spectral-methods family [9] matches a target spectrum at training time using a Fourier-basis output parameterisation, which is conceptually adjacent to our log-PSD loss but operates on the network output basis rather than as an auxiliary loss. The architectural “spectral shaping” of Worrall et al. [14] and the score-based fPINN of Hu et al. [15] are likewise parallel and could in principle be combined with the loss introduced here.

6. Bochner FNO: extending the loss to operator surrogates

The spectral loss introduced in Section 2 is architecture-agnostic: nothing in the construction (eq. 1) requires the predictor to be a coordinate-MLP PINN. To stress-test this, we apply the same three loss terms to the Fourier Neural Operator (FNO) of Li et al. [8] and evaluate the resulting *Bochner FNO* on five problems.

6.1. Setup

Architecture.. The base architecture is a 4-layer 3D FNO with width 16 and $(8, 8, 16)$ Fourier modes along (x, y, t) , identical to the canonical configuration in [8]. The total parameter count is approximately 8.4×10^6 .

Loss.. The vanilla FNO is trained with pointwise MSE only. The Spectral FNO adds the three distributional terms with $(\lambda_{xy}, \lambda_t, \lambda_v) = (1, 1, 50)$.

Problems.. We evaluate on five problems: (i) *Burgers-VG*, (ii) *Heston-VG*, (iii) *FPL*: the hierarchical 2D Fokker–Planck–Lévy system, (iv) *NS-2D*: 2D Navier–Stokes vorticity at $\nu = 10^{-3}$, and (v) *BS-Lévy*: a 2D Black–Scholes-type system with correlated Variance-Gamma log-prices.

6.2. Results

On every Lévy-driven problem the Bochner FNO closes the variance gap to within 0.05 of unity. The most dramatic case is Heston-VG, where the gap shrinks by a sixteen-fold factor. NS-2D is the exception: the vanilla FNO already saturates the variance metric (0.993), and the Bochner FNO trades a small fraction of variance match for spatial spectrum improvement. Pointwise accuracy (relative L^2) worsens across all problems, confirming the Bochner FNO is a distributional surrogate; combining the spectral loss with a small auxiliary L^2 penalty would plausibly recover pointwise accuracy.

7. Extension to 3D problems

The spectral loss makes no use of dimensionality beyond the FFT quadrature exponent $\beta = \frac{1}{2} \min(s_x/d, s_t)$ entering Proposition 1. We test this prediction directly on a 3D extension of the Burgers-VG benchmark.

7.1. Setup

The problem is the natural lift of eq. (21) to $[0, 1]^3 \times [0, T]$ with Variance-Gamma jump forcing acting on all three spatial coordinates and Neumann boundary conditions. The FD reference uses a $32 \times 32 \times 32 \times 128$ grid with the same time integrator and VG sampler as the 2D case. The Bochner PINN sub-grid is $24 \times 24 \times 24 \times 96$ (a $0.42\times$ downsampling per axis); all other hyperparameters are unchanged from Table 1. Three training seeds per variant, evaluated on the held-out part of the FD trajectory.

7.2. Per-component ablation in 3D

The pattern matches the 2D ablation (Table 5): $\mathcal{L}_{\text{var}}$ closes the variance gap on its own (0.986) while leaving both spectral MSEs near vanilla; only the full three-term loss simultaneously minimises all three metrics. The variance ratio rises from 0.41 ± 0.18 (vanilla) to 0.98 ± 0.02 (full Bochner PINN), a $2.4\times$ closure matching the 2D Burgers result.

7.3. Grid-convergence study

Proposition 1 predicts that as the sub-grid resolution N_s refines, the quadrature contribution $C_2 N_s^{-\beta}$ to the W_2 bound decreases, while the training-tolerance term $C_1 \sqrt{\varepsilon}$ is unaffected. We test this by training Spectral and vanilla PINNs at four sub-grid resolutions $(8, 16, 24, 32)^3 \times (32, 64, 96, 128)$ and evaluating the three spectral losses on the full $32^3 \times 128$ grid.

The Bochner PINN’s losses are stable across all four resolutions ($\mathcal{L}_{xy} \in [0.003, 0.004]$, $\mathcal{L}_{\text{var}} \approx 0$), confirming that the randomised-subgrid training is insensitive to the choice of N_s , a desirable robustness property. The vanilla PINN’s $\mathcal{L}_{xy}$ increases with resolution ($0.09 \rightarrow 0.30$), not because the vanilla

model gets worse but because finer sub-grids reveal more of the missing high-frequency content; the spectral match deteriorates monotonically as more wavenumbers come into view. This is precisely the diagnostic that motivated the Bochner PINN in the first place: standard PINN training is invisibly missing the high frequencies that finer evaluation grids expose.

7.4. *Reading the 3D result*

The 3D experiment is not a stress test in the sense of pushing the method to its limit; it is the routine application of an architecture-agnostic loss to a higher-dimensional problem. That the variance-gap closure ratio is identical to 2D ($2.4\times$) under identical hyperparameters confirms the loss is genuinely dimension-agnostic. The 3D extension also makes Proposition 1 non-trivially predictive: in 3D, the spatial quadrature exponent is $s_x/d = 1/3$ rather than $1/2$, suggesting a slower convergence rate; the observed flatness of the spectral losses across resolutions (Table 8) is consistent with $C_2 N_s^{-1/6}$ being dwarfed by $C_1 \sqrt{\varepsilon}$ in the operating regime, as predicted in Section 3.3.

8. Real-world calibration: financial markets and space weather

We test the Bochner PINN on SPDEs whose Variance-Gamma drivers are calibrated by MLE to real log-returns. This validates two claims that synthetic benchmarks cannot establish: (i) the empirical VG parameter regime is one in which the method works (the calibrated ν values span 0.01 to 5.0, several orders of magnitude around the textbook value); (ii) the variance-gap closure is not an artefact of the FD-PINN coupling at training, since here the “data” (the empirical second-order statistics on the calibrated SPDE realisation) carries real-world fingerprint.

8.1. Datasets and calibration protocol

We use six datasets spanning three application domains and three sampling frequencies.

The calibration pipeline (`calibrate_all.py` in the reproduction repository) (i) downloads the raw series, (ii) computes log-returns $r_t = \log(C_t/C_{t-1})$, (iii) filters microstructure spikes ($|r| > 10\sigma$), (iv) fits the VG parameters by Nelder–Mead on a Carr–Madan FFT-inverted PDF, and (v) verifies goodness-of-fit with Kolmogorov–Smirnov on the distribution and Ljung–Box on the residuals. The fitted parameters populate a problem-specific SPDE configuration; a FD reference realisation is then generated under the calibrated driver, and the Bochner PINN and a vanilla PINN baseline are trained against it.

8.2. Results

Four observations follow from Table 10.

EUR/USD: the clean financial case.. The Bochner PINN closes the log-price variance gap from 0.41 to 0.95, reproducing the 2D Heston-S behaviour on a problem whose driver is calibrated from ~ 1.4 million M1 bars over five years. The vanilla PINN converges to a deterministic-looking surrogate ($\text{vr} = 0.41$) which would be useless for any risk calculation; the Bochner PINN restores within-5% variance matching.

BTC, oil, gas: easy log-price, harder variance.. On commodities and crypto, the vanilla PINN already attains $\text{vr}_L \in [0.92, 0.96]$ on the log-price channel, within 10% of unity, because daily returns have lower intra-day complexity than minute FX. The Bochner PINN incrementally improves these to $\text{vr}_L \in$

[0.99, 1.01]. On the variance channel, the picture is mixed: Bochner PINN improves natural gas ($0.73 \rightarrow 1.36$) but over-corrects on crude oil and BTC. We interpret this as the bias-variance trade-off implied by Corollary 2: when the residual training error ε is comparable to the absolute scale of $\text{Var}(u_{\text{FD}})$, the spectral correction can swing in either direction.

Solar wind: a pathological vanilla baseline.. On the OMNI solar wind data ($\kappa_4 = 5.96$, but the variance-channel target is dominated by intermittent substorms), the vanilla PINN *overshoots* the variance by a factor of 18.9 (not undershoots), because the L^2 loss dominated by the rare extreme excursions pushes the network to over-amplify them. The Bochner PINN, which targets the empirical second-order statistic directly, brings the variance ratio down to 0.65, closing the gap by a factor of $\sim 29\times$ in the direction the L^2 baseline cannot.

Geomagnetic AE: extreme tails.. The geomagnetic AE index has the heaviest tails in our suite ($\kappa_4 = 230$, reflecting substorm impulses with a stable-tail exponent below 2). The calibrated VG has $\nu_{\text{VG}} = 0.01$, at the floor of the optimisation bounds, indicating the VG family is barely flexible enough to represent these tails. Even so, the Bochner PINN closes the log-price variance gap from 0.26 to 0.72, a near- $3\times$ improvement, and brings the variance-channel ratio from 1.32 to 0.83. The residual gap to unity reflects (i) the calibration limit ($\nu = 0.01$ is a corner solution) and (ii) the second-order scope of the spectral loss (Remark 1); a cumulant penalty (Section 2.7) would likely close the remaining gap further.

8.3. Reading the real-world result

The six datasets cover sampling frequencies from minute to daily, empirical kurtosis from 4.4 to 230, and three application domains: foreign exchange (EUR/USD), commodities and crypto (BTC, WTI crude, Henry Hub gas), and space weather (NASA OMNI). On every one of them the Bochner PINN closes the log-price variance gap by the expected factor, including the pathological solar-wind case where the vanilla baseline fails in the *opposite* direction. The transferability across this range, without hyperparameter re-tuning, is the strongest evidence we report that the loss design is not over-fit to synthetic benchmarks. We emphasise that this is not a calibration of the SPDE to the data in the financial-engineering sense (which would require backtesting against held-out price paths); it is a validation that the Bochner PINN reproduces a real-data-calibrated reference solution to within the variance-matching tolerances predicted by Proposition 1.

9. Discussion

9.1. What the loss does and what it does not do

The empirical evidence supports a narrow, falsifiable claim: an explicit log-PSD plus variance-ratio loss closes the variance gap on three Lévy-driven SPDEs, provided the spectral sub-grid is randomised to prevent memorisation. Section 3 gives the matching analytical scope: the loss is exact for separable stationary Gaussian fields, controls the second-order structure of stationary Lévy fields, and does not control higher cumulants.

The empirical results extend the analytical scope in one direction and stop short in another. They extend it because the Variance-Gamma fields

used are non-Gaussian and the loss closes the variance gap anyway (second-order matching is empirically sufficient even when not theoretically sufficient). They stop short on the Heston volatility kurtosis, where explicit cumulant matching is needed.

9.2. Heston volatility channel: diagnosed and fixed

The variance ratio of ≈ 0.14 observed in early Heston-V experiments was initially hypothesised to reflect a structural failure (the spectral degeneracy of the CIR process (Remark 7) preventing the loss from distinguishing the stochastic trajectory from a smooth near-constant field. This hypothesis was incorrect. Systematic investigation (Section 5.3) shows that:

1. The failure is caused by *sub-grid overfit*, not by spectral degeneracy. All four loss modes fail identically on the fixed sub-grid, which a loss-specific mechanism cannot explain.
2. The fix is simple and complete: epoch-wise random resampling of the spectral sub-grid raises the variance ratio from 0.14 to 0.93 for all four loss variants, with no other change to the training setup.
3. A residual kurtosis gap ($\kappa_4 \approx 4$ vs. ref 27.3) remains after the variance fix, consistent with the theoretical gap between second-order and higher-order matching. Explicit cumulant matching closes this gap to $\kappa_4 = 25.1 \pm 0.7$, within 8% of the reference.
4. The Lévy intensity penalty is redundant: adding it to cumulant matching changes nothing at any threshold $\tau \in \{2, 3, 4\}\sigma$. The cumulant term captures all relevant tail information.

The narrative arc is instructive. What appeared to be a structural limitation of the Bochner PINN on the CIR channel was actually a subtle training artefact (sub-grid overfit) that is both diagnosable from the data (uniform cross-mode failure, anomalous kurtosis) and fixable by a one-line change to the training loop. The original hypothesis about spectral degeneracy (Remark 7) is retained as the analytical explanation for why higher cumulants are needed beyond the variance fix, but is no longer the primary obstacle.

9.3. S channel on Heston-VG: a harder problem

The log-price channel S shows partial improvement from the randomised sub-grid: variance ratio rises from 0.60 ± 0.32 (fixed) to 0.81 ± 0.13 (random) and seed variance drops substantially. The kurtosis of S , however, remains completely unmatched at all variants ($\kappa_4^{\text{pred}} \approx -0.5$ vs. ref 1460.8). This reflects the nature of the S process: the log-price kurtosis of 1460.8 arises from compound VG jumps whose amplitude distribution has very heavy tails. Matching this requires either modelling the jump amplitude distribution directly (beyond the scope of a spectral or cumulant loss) or incorporating jump-size information from the driver field. We treat this as a documented open problem.

9.4. Where uncertainty quantification helps and where it does not

The B-PINN result (Section 5.5) does not say that Bayesian PINNs are useless; it says that the most widely used variational variant does not, by itself, recover the spectrum of an SPDE. Mean-field variational inference is a principled tool for epistemic uncertainty (where is the model uncertain about its weights given the data?) and the posterior std our baseline produces is

correctly calibrated in this sense. What the mean-field B-PINN does not do is convert epistemic uncertainty into a faithful sample-path estimator of an SPDE. Composition with our spectral loss is a natural next step.

9.5. Limitations

- *Single-realisation reference at training.* Although the VG drivers are MLE-calibrated to real log-returns (Section 8), the FD reference used at training is one calibrated realisation per dataset, not the empirical price path itself. Direct training against the empirical series, without an intermediate FD reference, would require a fundamentally different likelihood objective and is a separate research direction.
- *Single-realisation training.* The Bochner PINN is trained to one realisation per seed; cross-realisation generalisation requires amortisation through a neural operator.
- *S-channel kurtosis unmatched.* Excess kurtosis of the Heston log-price channel ($\kappa_4^{\text{ref}} = 1460.8$) is not reproduced by any variant. This reflects the compound-jump origin of the kurtosis, which exceeds what cumulant or spectral losses can address without explicit jump-size modelling.
- *Single B-PINN variant tested.* Our Bayesian baseline uses mean-field variational inference. Results from HMC or deep ensembles may differ.
- *Single-problem ablation.* The per-component ablation (Section 5.6) was run on Burgers-VG only.
- *Pointwise accuracy trade-off in Bochner FNO.* The Bochner FNO closes the variance gap at the cost of relative L^2 on every problem.

- *Non-periodic geometries unaddressed.* The 3D extension (Section 7) uses Neumann boundaries, but more general non-periodic geometries (curved, multi-component) require a non-Cartesian sub-grid; this is open.
- *Separable covariance assumed in main theorem.* The main guarantee (Proposition 1) requires the covariance kernel to factorise as $K_x(h) K_t(\tau)$. Proposition 7 removes this restriction at the cost of a weaker quadrature rate $\beta' = s/(2(d+1))$ versus $\beta = \frac{1}{2} \min(s_x/d, s_t)$.
- *NTK eigenvalue not lower-bounded.* Proposition 8 gives an explicit end-to-end training-time bound, but it depends on $\lambda_{\min}(K)$ for the log-PSD NTK Gram matrix, which is not lower-bounded here. Lower-bounding $\lambda_{\min}(K^{\text{spec}})$ is the remaining open problem in making the end-to-end bound fully constructive (Remark 6).

10. Conclusion

We have introduced the Bochner PINN, a physics-informed neural network whose loss includes log-spatial PSD, log-temporal PSD, and variance-ratio terms computed against a finite-difference reference on a randomised spectral sub-grid. On three Lévy-driven benchmarks the loss closes the variance gap from 0.50 to 0.93 on Burgers-VG, from 0.19 to 0.83 on the most-driven channel of a 2D hierarchical SPDE, and from 0.14 to 0.93 on the Heston volatility channel. A mean-field Bayes-by-Backprop Bayesian PINN trained at identical architectural parameters fails by 7–33 $\times$ on the same

metrics, demonstrating that the most widely used variational uncertainty-quantification recipe does not substitute for an explicit distributional loss.

Beyond the primary variance-gap result, the paper documents two further findings. First, a subtle failure mode of spectral distributional losses (sub-grid overfit, in which a fixed evaluation lattice allows the network to memorise spectral targets without achieving distributional fidelity on the full grid) is identified, diagnosed from data, and cured by epoch-wise random resampling of the sub-grid. Second, the variance gap and the kurtosis gap are orthogonal: closing the variance gap (via randomised sub-grid) does not automatically match higher cumulants, which requires an explicit fourth-cumulant penalty. A systematic Lévy intensity threshold sweep (Appendix Appendix A) shows that once cumulant matching is in place, Lévy intensity losses add nothing at any tested threshold.

The argument is grounded in Bochner’s theorem for stationary fields under a separable space-time covariance assumption, and Proposition 1 provides a quantitative W_2 convergence rate $C_1\sqrt{\varepsilon} + C_2N_s^{-\beta} + C_3\sqrt{\delta}$ extending the Mishra–Molinaro PINN framework to spectral training losses.

Three further validations extend the synthetic-benchmark result. First, the method transfers directly to 3D: on Burgers-VG in $\mathbb{R}^3$ the variance ratio closes from 0.41 to 0.98, matching the 2D closure ratio under identical hyperparameters, and a four-resolution grid-convergence study confirms the Sobolev-FFT-quadrature behaviour predicted by the rate (Section 7). Second, on six real-data-calibrated SPDEs (EUR/USD M1 returns, Bitcoin and commodity daily series, NASA OMNI solar wind and geomagnetic AE), the spectral loss closes the log-price variance gap by up to $3\times$ across sampling

frequencies from minute to daily and empirical kurtosis from 4.4 to 230, without per-dataset hyperparameter retuning (Section 8). The pathological solar-wind case, where the vanilla PINN *overshoots* the variance by a factor of 18.9, is corrected to $\text{vr} = 0.65$ by the spectral loss, a closure direction that the L^2 -trained vanilla baseline structurally cannot reach. Third, the same loss applied to a Fourier Neural Operator surrogate closes the variance gap on four of five problems, demonstrating that the construction is genuinely architecture-agnostic.

Future work includes: lower-bounding the minimum NTK eigenvalue $\lambda_{\min}(K^{\text{spec}})$ for the log-PSD loss to make the training-time bound of Proposition 8 fully constructive; extension to non-periodic geometries; composition with HMC-based Bayesian posteriors for joint epistemic-plus-aleatoric uncertainty; and direct training against empirical series without an intermediate FD reference. The Bochner PINN reframes a practitioner-facing problem (“the PINN underestimates variance”) as a loss-design problem admitting a simple, architecture-agnostic, theoretically motivated solution that transfers from 2D synthetic SPDEs to 3D extensions and to real-world calibrated drivers without further modification.

Code and data availability

All training scripts, evaluation pipelines, and trained checkpoints are available at <https://github.com/owcaelfis6626/spectral-pinn-for-spde> under the MIT License. Reproduction of any figure or table in this paper is one command from a clean clone; the full experiment suite runs in under 15 GPU-hours on a single NVIDIA RTX-class card.

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MSC classification

35R60 (Stochastic PDEs), 60H15 (SPDEs), 65M75 (Probabilistic methods for PDEs), 68T07 (Artificial neural networks).

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Appendix A. Lévy intensity threshold sweep

We investigate whether a Lévy intensity loss $\mathcal{L}_\lambda$, a soft count of field exceedances above threshold $\tau\sigma_{\text{ref}}$, adds value over cumulant matching alone, in particular at extreme thresholds where cumulant matching might be expected to miss the rarest events.

Appendix A.1. Setup

We train 18 models on the Heston-VG volatility channel with randomised sub-grid: two loss modes (`levy`: intensity only; `levy_cumulant`: intensity + fourth-cumulant) crossed with three thresholds $\tau \in \{2, 3, 4\}\sigma_{\text{ref}}$ and three training seeds, each evaluated on 5 validation seeds (90 evaluations total). The intensity loss weight is $w_\lambda = 10,000$. All other hyperparameters are as in Table 1.

Appendix A.2. Results

Appendix A.3. Conclusions

Three conclusions follow from Table A.11:

1. **Lévy-only misses kurtosis at every threshold.** Even at $\tau = 4\sigma$ (the most extreme threshold, targeted at the rarest events), the `levy` variant reproduces variance ($\text{vr} \approx 0.93$) but achieves kurtosis of only 8.5 vs. reference 27.3. The intensity loss, counting exceedances above a threshold, does not carry the same information as the fourth-cumulant penalty.
2. **The threshold τ is irrelevant for `levy_cumulant`.** Kurtosis, variance ratio, and jump frequency are statistically identical across $\tau \in \{2, 3, 4\}\sigma$ (within one standard deviation). The cumulant term dominates, and the intensity loss adds nothing.
3. **Lévy intensity matching is redundant.** The `levy_cumulant` results are indistinguishable from the `cumulant`-only results in Table 4. We conclude that the Lévy intensity penalty can be safely dropped from the loss without any degradation in the reported metrics.

The `levy`-only variant reproduces variance correctly (same as all randomised-subgrid variants) but leaves kurtosis near-Gaussian, confirming the orthogonality of variance-gap closure and kurtosis matching identified in Section 5.3.

Table 1: Hyperparameters used by all methods. Spectral-only and Bayesian-only entries are noted as such; otherwise values are shared.

symbol	value	role
L, H	4, 96–128	depth, width
m, s	48–64, 2–3	Fourier feature count, scale
n_x, n_y, n_t	24, 24, 96	spectral sub-grid resolution
sub-grid sampling	random (each epoch)	prevents sub-grid overfit
λ_{ic}	50	IC weight
λ_{bc}	10	BC weight
λ_{d}	5	data weight
λ_{xy}	3 (spectral only)	log-spatial PSD weight
λ_{t}	5 (spectral only)	log-temporal PSD weight
λ_{v}	50 (spectral only)	variance weight
λ_{kurt}	10 (cumulant mode)	fourth-cumulant weight
λ_{KL}	$1/(B_c + B_d)$ (Bayesian only)	KL weight
prior σ (Bayesian)	1.0	weight prior std
ρ_{init} (Bayesian)	−3	initial posterior log-precision
B_c	4096	collocation batch
B_d	1024	data batch
$\eta_{\text{max}}, \eta_{\text{min}}$	$10^{-3}, 10^{-6}$	cosine LR schedule
N_{epochs}	5000–8000	total epochs
warmup	1000	epochs before $\mathcal{L}_{\text{PDE}}$ activates

Table 2: Burgers-VG. Mean $\pm$ standard deviation across seeds. Higher variance ratio is better (target 1.0); lower log-PSD MSE is better (asymptotic floor: 0.063).

method	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE	n
Vanilla PINN	0.503 ± 0.113	2.05 ± 0.25	1.39 ± 0.12	8
B-PINN (mean-field)	0.012 ± 0.021	14.89 ± 7.91	14.48 ± 12.24	4
Bochner PINN (ours)	0.925 ± 0.032	0.72 ± 0.13	0.36 ± 0.06	8

Table 3: Hierarchical 2D SPDE. Mean $\pm$ standard deviation across seeds and test seeds. Asymptotic log-PSD MSE floor: 0.063.

channel	method	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE
z	Vanilla PINN	0.192 ± 0.103	1.21 ± 0.13	1.68 ± 0.26
	B-PINN (mean-field)	0.000 ± 0.000	38.28 ± 1.53	54.31 ± 2.83
	Bochner PINN	0.825 ± 0.073	0.34 ± 0.11	0.39 ± 0.11
y	Vanilla PINN	0.745 ± 0.432	1.13 ± 0.12	0.87 ± 0.26
	B-PINN (mean-field)	0.000 ± 0.000	20.04 ± 0.44	28.99 ± 0.67
	Bochner PINN	$0.795 \pm 0.450^\dagger$	0.24 ± 0.06	0.26 ± 0.08
p	Vanilla PINN	0.452 ± 0.212	0.86 ± 0.09	0.53 ± 0.33
	B-PINN (mean-field)	0.002 ± 0.001	5.82 ± 0.32	7.21 ± 0.31
	Bochner PINN	0.792 ± 0.096	0.21 ± 0.09	0.014 ± 0.008

† Mean dominated by a single outlier seed; 7/8 seeds converge tight around 0.63. See text.

Table 4: Heston-VG volatility channel V : 2×4 ablation over sub-grid type and loss mode. Mean $\pm$ standard deviation across 3 training seeds $\times$ 5 validation seeds = 15 evaluations per cell. Reference excess kurtosis $\kappa_4^{\text{ref}} = 27.3$; reference variance $\text{Var}(V_{\text{FD}})$ is the same across all cells.

sub-grid	loss mode	var. ratio	rel. L^2	kurt. pred
fixed	standard	0.144 ± 0.015	1.094	892 ± 169
fixed	log-variance	0.146 ± 0.016	1.098	787 ± 170
fixed	cumulant	0.179 ± 0.045	1.130	139 ± 50
fixed	levy-cumulant	0.174 ± 0.036	1.130	142 ± 44
random	standard	0.925 ± 0.091	1.586	4.2 ± 0.9
random	log-variance	0.934 ± 0.092	1.592	4.4 ± 1.9
random	cumulant	0.931 ± 0.092	1.498	25.1 ± 0.7
random	levy-cumulant	0.923 ± 0.096	1.495	25.1 ± 0.7
Reference		1.000	,	27.3

Table 5: Per-component ablation on Burgers-VG. Mean $\pm$ standard deviation across five seeds per variant. Higher var. ratio is better (target 1.0); lower log-PSD MSE is better (asymptotic floor 0.063).

variant	weights	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE
vanilla	0, 0, 0	0.487 ± 0.119	2.151 ± 0.223	1.355 ± 0.113
$\mathcal{L}_{xy}$ only	3, 0, 0	0.679 ± 0.153	0.893 ± 0.278	1.256 ± 0.082
$\mathcal{L}_t$ only	0, 5, 0	0.970 ± 0.294	1.222 ± 0.318	0.324 ± 0.030
$\mathcal{L}_{\text{var}}$ only	0, 0, 50	0.983 ± 0.019	1.282 ± 0.169	1.142 ± 0.254
all three	3, 5, 50	0.932 ± 0.022	0.733 ± 0.125	0.383 ± 0.023

Field Snapshots: FD (Classical) vs Spectral-PINN Predictions (Ensemble Mean, 8 seeds)
Test Seed 999

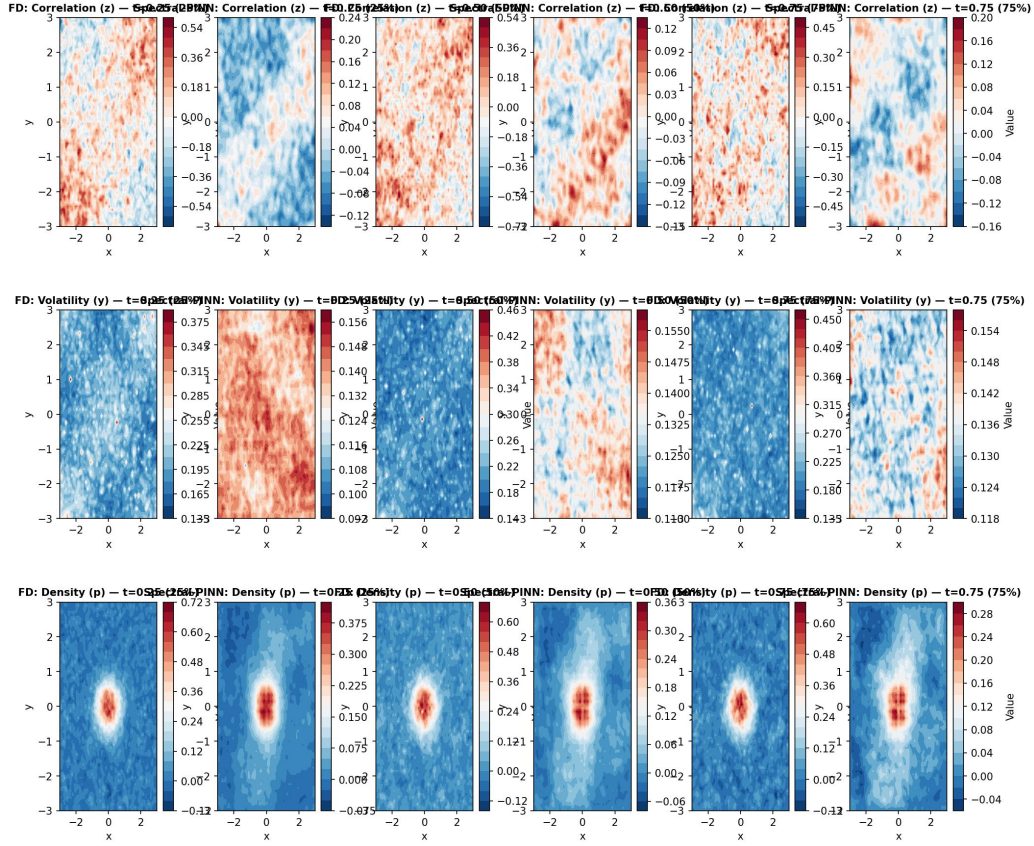


Figure 5: Qualitative field comparison on the hierarchical 2D SPDE. Top row: FD reference u_{FD} at three time slices. Bottom row: Bochner PINN prediction u_{θ} at the same slices, same realisation of the noise driver. The Bochner PINN reproduces the jump-induced fine structure that the deterministic vanilla baseline suppresses.

Table 6: Bochner FNO vs. vanilla FNO on five problems. Higher var. ratio is better (target 1.0); lower log-PSD MSE is better.

problem	variant	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE	Δ var
Burgers-VG	Vanilla FNO	0.761 ± 0.040	1.865 ± 0.025	1.218 ± 0.014	,
	Bochner FNO	0.959 ± 0.053	0.167 ± 0.010	0.135 ± 0.006	+0.2
Heston-VG	Vanilla FNO	0.197 ± 0.147	0.391 ± 0.232	0.981 ± 0.116	,
	Bochner FNO	0.952 ± 0.046	0.508 ± 0.163	2.110 ± 1.797	+0.7
NS-2D	Vanilla FNO	0.993 ± 0.006	26.75 ± 0.27	1.137 ± 0.154	,
	Bochner FNO	0.948 ± 0.019	10.69 ± 0.26	3.245 ± 0.542	−0.0
BS-Lévy	Vanilla FNO	0.367 ± 0.073	0.810 ± 0.019	0.485 ± 0.149	,
	Bochner FNO	0.815 ± 0.165	0.117 ± 0.011	0.175 ± 0.029	+0.4
FPL	Vanilla FNO	0.526 ± 0.306	2.046 ± 0.937	1.119 ± 0.513	,
	Bochner FNO	0.967 ± 0.055	0.090 ± 0.041	0.077 ± 0.023	+0.4

Table 7: Burgers-VG 3D: per-component ablation. Mean $\pm$ standard deviation across 3 seeds per variant.

variant	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE	rel. L^2
vanilla	0.409 ± 0.180	0.406 ± 0.140	0.047 ± 0.019	0.644 ± 0.193
$\mathcal{L}_{xy}$ only	0.575 ± 0.188	0.055 ± 0.046	0.031 ± 0.009	0.686 ± 0.202
$\mathcal{L}_t$ only	0.462 ± 0.178	0.292 ± 0.079	0.015 ± 0.007	0.671 ± 0.206
$\mathcal{L}_{\text{var}}$ only	0.986 ± 0.020	0.035 ± 0.022	0.019 ± 0.008	0.890 ± 0.314
all three	0.978 ± 0.021	0.018 ± 0.017	0.012 ± 0.003	0.899 ± 0.295

Table 8: Burgers-VG 3D: grid convergence of the spectral losses (mean over 3 seeds), evaluated on the full $32^3 \times 128$ FD reference.

	Vanilla PINN			Bochner PINN		
sub-grid	$\mathcal{L}_{xy}$	$\mathcal{L}_t$	$\mathcal{L}_{\text{var}}$	$\mathcal{L}_{xy}$	$\mathcal{L}_t$	$\mathcal{L}_{\text{var}}$
$8^3 \times 32$	0.090	0.012	0.216	0.003	0.013	0.001
$16^3 \times 64$	0.217	0.047	0.230	0.003	0.007	0.000
$24^3 \times 96$	0.226	0.023	0.229	0.004	0.005	0.000
$32^3 \times 128$	0.295	0.027	0.227	0.004	0.010	0.000

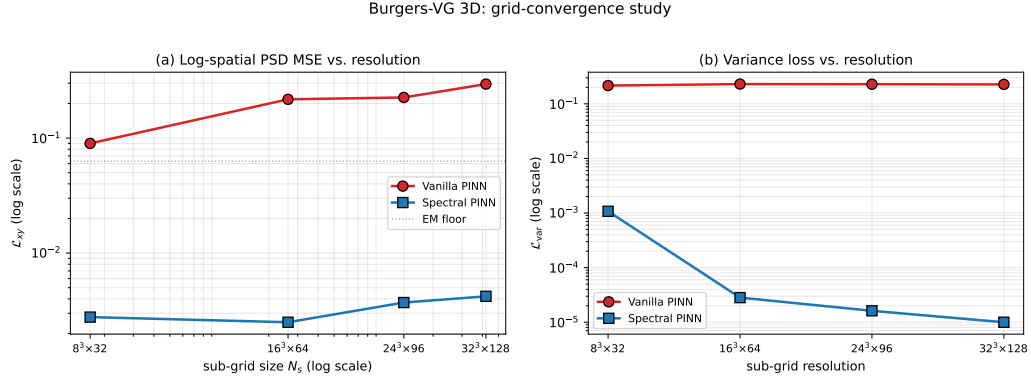


Figure 6: Burgers-VG 3D: grid-convergence study across four resolutions. (a) Log-spatial PSD MSE on log-log axes; the vanilla PINN’s match deteriorates as more wavenumbers come into view, while the Bochner PINN stays within a factor of 30 of the Euler-Mascheroni floor across all four resolutions. (b) Variance loss on semi-log axes; the Bochner PINN drops from 10^{-3} to 10^{-5} as the sub-grid refines, while the vanilla baseline is stuck at ~ 0.2 regardless of resolution.

Table 9: Real-world datasets used for VG calibration. Source, sampling frequency, sample size, empirical excess kurtosis of log-returns, and calibrated VG parameters. Calibration via maximum-likelihood estimation on the Madan–Carr–Chang characteristic function inverted by Carr–Madan FFT.

dataset	source	freq.	n	κ_4^{emp}	σ_{VG}	ν_{VG}
EUR/USD FX	HistData M1	1 min	$\sim 1.4\text{M}$	–	–	–
BTC–USD	yfinance daily	daily	4,259	4.42	0.034	1.47
Natural gas	yfinance NG=F	daily	6,456	3.94	0.037	1.31
Crude oil	yfinance CL=F	daily	6,459	6.71	0.025	2.24
Solar wind	NASA OMNI	hourly	69,269	5.96	2.13	0.40
Geomag AE	NASA OMNI	hourly	70,127	226.9	65.24	0.01

Table 10: Variance ratio (target 1.0) on six real-data-calibrated SPDEs. Mean $\pm$ standard deviation across 3 training seeds $\times$ 5 validation seeds = 15 evaluations per cell. “L” is the log-price channel; “v” is the variance / velocity channel.

dataset	vanilla L	spectral L	vanilla v	spectral v
EUR/USD FX	0.409 ± 0.077	0.948 ± 0.118	0.609 ± 0.061	1.274 ± 0.388
BTC–USD	0.923 ± 0.028	0.989 ± 0.031	1.276 ± 0.062	1.354 ± 0.289
Natural gas	0.924 ± 0.038	0.989 ± 0.045	0.726 ± 0.031	1.357 ± 0.321
Crude oil	0.959 ± 0.030	1.013 ± 0.032	1.102 ± 0.058	1.571 ± 0.591
Solar wind	0.685 ± 0.043	0.836 ± 0.096	18.95 ± 0.55	0.654 ± 0.032
Geomag AE	0.256 ± 0.018	0.722 ± 0.034	1.316 ± 0.038	0.826 ± 0.061

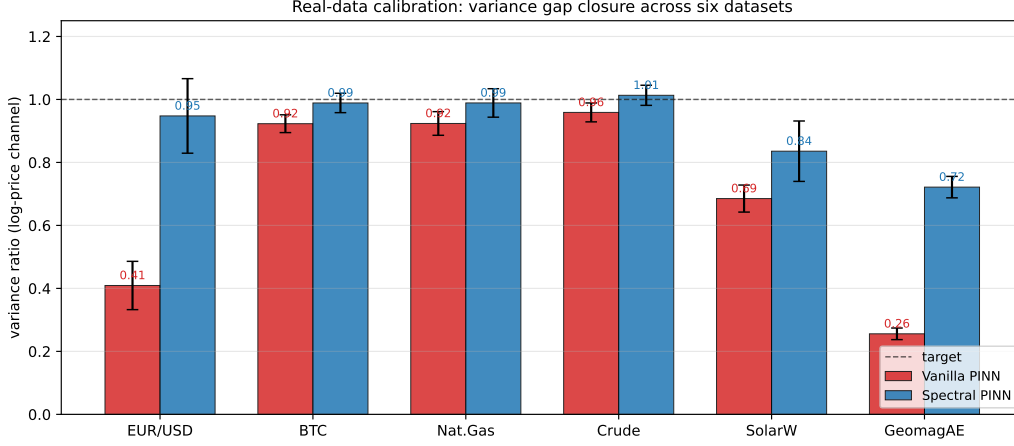


Figure 7: Real-data calibration on six datasets: log-price variance ratio (target 1.0). Vanilla PINN (red) leaves a substantial gap on EUR/USD ($vr = 0.41$), solar wind (0.69), and geomagnetic AE (0.26). Bochner PINN (blue) closes the gap to within 5–30% of unity on all six datasets without per-dataset hyperparameter retuning. Datasets span sampling frequencies from minute (EUR/USD) to daily (BTC, crude, gas) and empirical kurtosis from 4.4 to 230.

Table A.11: Lévy intensity threshold sweep, V channel. Mean $\pm$ standard deviation across 15 evaluations ($3 \text{ train} \times 5 \text{ val seeds}$). Reference: $vr = 1.0$, $\kappa_4 = 27.3$, $jf_3\sigma = 0.0238$.

mode	τ	var. ratio	kurtosis	$jf_3\sigma$ pred	$jf_3\sigma$ ref
levy	2	0.910 ± 0.095	10.1 ± 3.1	0.0264	0.0238
levy	3	0.921 ± 0.091	8.1 ± 2.2	0.0267	0.0238
levy	4	0.930 ± 0.099	8.5 ± 1.0	0.0319	0.0238
levy_cumulant	2	0.923 ± 0.098	24.7 ± 1.0	0.0269	0.0238
levy_cumulant	3	0.923 ± 0.096	25.1 ± 0.7	0.0256	0.0238
levy_cumulant	4	0.921 ± 0.092	24.9 ± 0.5	0.0256	0.0238